

MODELING URBAN CRIME RISK AND ARREST PROBABILITY USING SPATIO-TEMPORAL MACHINE LEARNING TECHNIQUES: EVIDENCE FROM CHICAGO

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ABSTRACT

This study presents a machine learning framework for determining whether a reported crime will result in an arrest by integrating crime records with socio-economic, weather, and public transportation mobility data. Using the Chicago Crimes dataset (2001–2023), extensive data preprocessing, feature engineering, and exploratory data analysis were conducted to identify spatial and temporal crime patterns. Multiple machine learning models were evaluated using the original dataset and class-balancing techniques, including random undersampling and oversampling. Model performance was assessed using Accuracy, Precision, Recall, F1-score, Receiver Operating Characteristic–Area Under the Curve (ROC-AUC), and Precision–Recall Area Under the Curve (PR-AUC). Experimental results demonstrate that balancing techniques significantly improve arrest prediction, with tree-based ensemble models achieving the strongest predictive performance; a Random Forest classifier trained on the randomly over-sampled training data was selected as the final model, achieving the best overall balance of F1-score, ROC-AUC, and PR-AUC. The findings highlight the value of integrating heterogeneous urban environmental data to support evidence-based policing, crime prevention, and public safety decision-making.

Key words: Arrest Prediction; Machine Learning; Spatio-Temporal Analysis; Crime Analytics; Urban Environmental Factors

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Contents

1	Introduction	1
2	Literature Review	4
2.1	Spatiotemporal Deep Learning for Crime Forecasting	4
2.2	Arrest Prediction and Decision Support	5
2.3	Multi-Source and Multimodal Data Integration	6
2.4	Traditional Machine Learning and Crime-Type-Specific Studies	6
2.5	Theoretical Foundations	7
3	Data Description and Exploratory Data Analysis	9
3.1	Dataset Description	9
3.2	Data Preprocessing and Integration	10
3.3	Feature Engineering and Feature Selection	13
3.4	Final Modeling Dataset and Variables	15
3.5	Exploratory Data Analysis	16
4	Methodology	21
4.1	Study Aim and Experimental Overview	21
4.2	Experimental Design	21
4.3	Models and Hyperparameter Tuning	22
4.4	Evaluation Metrics	25
5	Results and Discussion	28
5.1	Results by Sampling Strategy	28
5.2	Overall Model and Sampling Comparison	31
5.3	Discussion	37
5.3.1	Arrest Prediction Performance	37
5.3.2	Contribution of Predictor Groups	38
5.3.3	Class Imbalance	39
5.4	Limitations and Future Research	40
A	Supplementary Exploratory Data Analysis Figures	43

List of Tables

3.1	Predictor Variable Categories	16
4.1	Hyperparameter Configuration – Classical Machine Learning Models . . .	24
4.2	Hyperparameter Configuration – Neural Network Architectures	25
4.3	Confusion Matrix	26
5.1	Model Performance on the Original Imbalanced Training Data	29
5.2	Model Performance with Random Under-Sampling	30
5.3	Model Performance with Random Over-Sampling	31

List of Figures

3.1	Crimes and Arrest Rate per Year	16
3.2	Arrest Rate by Crime Type	17
3.3	Top Community Areas by Crime Volume	18
3.4	Socio-Economic Factors vs. Total Crimes	18
3.5	Crimes and Arrest Rate by Temperature Category	19
3.6	Correlation Heatmap	20
5.1	Average Performance of Different Sampling Methods	32
5.2	Top 10 Performing Models	33
5.3	F1-Score Comparison Across Sampling Methods	34
5.4	ROC-AUC Comparison of Machine Learning Models	35
5.5	Precision-Recall Curves of the Best-Performing Models	36
A.1	Crimes and Arrest Rate by Month	43
A.2	Top 15 Crime Types by Volume	43
A.3	Crimes and Arrest Rate by Hour	44
A.4	Crimes and Arrest Rate by Day of Week	44
A.5	Arrest Rate by Income and Unemployment Quintiles	44
A.6	Annual Crime Count and Transit Ridership Trends.	45
A.7	Pre-COVID-19, COVID-19, and Post-COVID-19 Crime and Mobility Comparison	45
B.1	Accuracy Comparison Across Sampling Methods	46
B.2	Precision Comparison Across Sampling Methods	47
B.3	Recall Comparison Across Sampling Methods	47

1. Introduction

Crime remains one of the most persistent challenges confronting modern cities, threatening public safety, undermining economic stability, and diminishing the overall quality of urban life [1]. This is not merely a theoretical concern: according to the FBI’s Crime Data Explorer, police nationwide cleared only 36.7% of reported violent crimes and just 12.1% of reported property crimes in 2022, meaning the majority of reported offences never result in an arrest or referral for prosecution [2]. Crime is influenced by the interaction of spatial, temporal, socio-economic, and environmental factors, making criminal behaviour inherently complex and difficult to predict [1, 3]. Traditional analytical and statistical approaches, which rely primarily on historical crime statistics and manual analysis, often struggle to capture these complex relationships and therefore have limited capability to accurately predict crime patterns and arrest outcomes [4, 3]. Consequently, there has been growing interest in adopting data-driven approaches capable of supporting intelligent, evidence-based decision-making through predictive analytics and machine learning [3, 5].

The rapid development of information technology has enabled government agencies to collect and maintain massive volumes of crime-related data [3, 4]. Publicly available datasets, such as the Chicago Crimes dataset released through the City of Chicago Open Data Portal, provide detailed information on crime type, location, time, and arrest status, offering valuable opportunities for researchers to study crime patterns using computational techniques [5, 6]. At the same time, advances in artificial intelligence, machine learning, and deep learning have transformed crime analysis from descriptive statistical reporting into predictive modelling capable of uncovering hidden patterns within complex datasets [3, 7, 8]. Numerous studies have shown that machine learning algorithms consistently outperform conventional statistical approaches across a range of crime prediction tasks, primarily because they can model nonlinear relationships among variables and learn complex decision boundaries from large volumes of data [5, 9, 8, 7].

Crime is inherently shaped by both spatial and temporal characteristics [9, 5, 10]. Spatial information describes where a crime occurs and includes features such as geographic coordinates, police districts, community areas, and location descriptions, while temporal information describes when a crime occurs and includes variables such as year, month, day of the week, date, and hour [9, 5]. Collectively, these features capture long-term trends, seasonal variation, daily activity patterns, and regional concentrations of offending

[5, 9, 11]. Previous studies have shown that incorporating both spatial and temporal information significantly improves predictive performance because criminal activity tends to cluster around specific locations and recur at particular times rather than occurring uniformly across space and time [5, 9, 10, 11].

In addition to spatial and temporal structure, recent research indicates that crime is also shaped by the broader urban environment, defined here as the combination of socio-economic conditions, weather, and human mobility patterns that characterize a neighborhood or city at a given time [12, 6, 13]. Socio-economic indicators, including income level, unemployment rate, poverty, and educational attainment, have been widely associated with crime occurrence and public safety [6]. Likewise, weather conditions such as temperature, rainfall, snowfall, and wind speed can influence human activity and, consequently, crime patterns [10]. Human mobility, represented through public transportation usage and daily travel behaviour, provides additional information about population movement and urban activity levels [13, 12]. Integrating these heterogeneous data sources allows predictive models to better represent real-world urban environments and improve crime prediction performance [6, 10, 12, 13].

Over the past decade, numerous machine learning and deep learning models have been proposed for crime prediction [3, 7]. Ensemble learning techniques, recurrent neural networks, graph neural networks, transformer-based neural networks, and spatio-temporal deep learning architectures have all demonstrated promising performance in forecasting crime occurrence and identifying crime hotspots [5, 9, 8, 7]. Studies employing Long Short-Term Memory (LSTM) networks, Spatial-Temporal Graph Convolutional Networks (ST-GCN), the Informer architecture, Temporal Delay Neural Networks, residual neural networks, and multimodal deep learning models have further demonstrated the importance of capturing both spatial and temporal dependencies within crime datasets [5, 9, 14, 15, 12, 16, 17]. Despite these advances, most existing studies focus primarily on predicting crime occurrence, crime hotspots, or crime categories, and comparatively little attention has been paid to arrest prediction—that is, predicting whether a reported crime will ultimately result in an arrest [5, 9, 3].

Predicting whether a reported crime is likely to result in an arrest carries direct practical value for law enforcement and the communities they serve [6]. Such predictions can help police departments prioritise investigative resources, identify crime types or neighbourhoods requiring additional attention, and support evidence-based policing strategies [6]. Despite its practical significance, arrest prediction remains comparatively underexplored, as most existing studies focus primarily on crime occurrence, crime hotspots, or crime categories while relying mainly on historical crime records and giving limited attention to broader urban contextual information such as socio-economic conditions, weather vari-

ables, and mobility patterns [5, 9].

Taken together, the literature reveals three main gaps that motivate the present study. First, most existing work concentrates on predicting crime occurrence, crime hotspots, or crime categories, with comparatively few studies directly modelling arrest outcomes [5, 9, 3, 6]. Second, existing arrest- and crime-prediction studies tend to rely mainly on historical crime records together with spatial and temporal variables, without systematically accounting for broader contextual information [6, 9, 5]. Third, limited research integrates socio-economic, weather, and public transportation mobility data within a single predictive framework, so the combined influence of these urban environmental factors on arrest outcomes remains poorly understood [6, 10, 12].

To address these gaps, this study develops a machine learning framework for predicting whether a reported crime will result in an arrest by integrating publicly available crime records from the City of Chicago with socio-economic indicators, weather observations, and public transportation mobility data. The study systematically investigates the influence of spatial, temporal, socio-economic, environmental, and mobility-related factors on arrest prediction while addressing the challenges associated with class imbalance. Multiple machine learning models are evaluated using the original dataset together with random under-sampling and random over-sampling strategies to identify the most effective approach for arrest prediction. Overall, this study makes three main contributions: it develops a comprehensive machine learning framework for arrest prediction; it demonstrates the value of integrating heterogeneous urban environmental data within a unified predictive model; and it provides a systematic comparison of class-balancing strategies for improving arrest prediction performance.

The remainder of this Major Research Paper is organized as follows. Chapter 2 reviews the existing literature on crime prediction and arrest prediction using machine learning and deep learning techniques. Chapter 3 describes the datasets used in this study and presents the data preprocessing procedures, feature engineering, and exploratory data analysis. Chapter 4 presents the methodology, including class-balancing techniques and the machine learning models employed in this research. Finally, Chapter 5 presents the experimental results, discusses the findings, outlines the limitations of the study, and suggests directions for future research.

2. Literature Review

Crime prediction has evolved considerably over the past decade, progressing from theory-driven criminological explanations to data-intensive machine learning and deep learning approaches. This chapter reviews the existing literature across five thematic areas: spatiotemporal deep learning architectures for crime forecasting, arrest prediction and law enforcement decision support, multi-source and multimodal data integration, traditional machine learning and crime-type-specific studies, and the theoretical foundations underpinning data-driven crime research. The literature identifies a consistent research gap that motivates the present study: the lack of a comprehensive machine learning approach that integrates crime, weather, mobility, and socio-economic data specifically for arrest outcome prediction while systematically evaluating both classical machine learning and deep learning models.

2.1 Spatiotemporal Deep Learning for Crime Forecasting

A dominant theme in recent literature combines graph-based spatial modelling with sequence-based temporal modelling to capture the dual nature of crime patterns. Han et al. integrated LSTM networks with ST-GCN to forecast theft crimes in Chicago, incorporating weather and holiday variables and demonstrating that joint spatiotemporal modelling outperformed single-dimension approaches [9]. Building on this architecture, Fan et al. replaced the LSTM component with an Informer-based encoder, leveraging its efficiency in processing long sequences to improve prediction across multiple crime types, including assault, robbery, theft, and criminal damage, using approximately 320,000 records from 22 Chicago police districts [5]. The Informer architecture, originally proposed by Zhou et al. for long-sequence time-series forecasting, introduced ProbSparse self-attention and generative decoding to overcome the computational limitations of standard Transformers and LSTMs. Since its introduction, the Informer has been adapted for several crime forecasting applications, including the work of Fan et al. [16, 5].

More recent studies have increasingly focused on transformer-based and hierarchical architectures. Riyazuddin et al. proposed a Hierarchical Spatiotemporal Transformer (HSTT) that integrated structured crime data, mobility signals, and contextual factors while also

addressing fairness, interpretability, and uncertainty estimation, aspects that were largely absent from earlier models [8]. Similarly, hybrid CNN-LSTM architectures have been explored for hotspot classification. A Spatial-First-Then-Temporal (SFTT) architecture evaluated across five U.S. cities demonstrated that learning spatial structure before temporal dynamics improved hotspot prediction accuracy compared with traditional machine learning baselines [7]. Earlier recurrent approaches, such as the Spatiotemporal Neural Network (STNN) applied to Portland crime data and the ST-ResNet framework originally developed for citywide crowd-flow prediction, established fundamental spatiotemporal modelling principles, including recent observations, periodic daily patterns, and long-term trends, that continue to influence crime prediction architectures [11, 17]. A ternarised adaptation of ST-ResNet was subsequently applied to real-time hourly crime intensity forecasting in Los Angeles, although the study was limited to a six-month observation period [18].

Collectively, these studies demonstrate that spatiotemporal deep learning models consistently outperform non-spatial or purely temporal approaches. However, most existing studies focus on predicting crime occurrence or hotspot locations rather than arrest outcomes, and only a limited number incorporate socio-economic or mobility-related variables alongside spatiotemporal information.

2.2 Arrest Prediction and Decision Support

In contrast to the extensive literature on crime occurrence forecasting, research specifically focused on arrest prediction remains limited. Sankul et al. conducted one of the few studies to directly model arrest likelihood using Random Forest and other classical machine learning algorithms applied to historical crime records enriched with spatial, temporal, and socio-economic features. Their findings demonstrated the potential of machine learning to support law enforcement decision-making and resource allocation [6]. However, the study relied exclusively on traditional machine learning techniques and did not incorporate spatiotemporal deep learning architectures or mobility-related information. Consequently, it remains unclear whether the performance improvements achieved by advanced deep learning models for crime occurrence prediction can be extended to arrest prediction.

2.3 Multi-Source and Multimodal Data Integration

A second major theme in the literature concerns the integration of heterogeneous external data sources beyond raw crime records. Wang et al. demonstrated that combining Chicago crime and census data with Point-of-Interest (POI) information and taxi mobility data substantially improved crime-rate estimation compared with using demographic variables alone, highlighting the significant contribution of human mobility patterns to crime prediction [13]. This concept was further extended through the Cross-Type and Spatiotemporal Correlations (CCC) framework, which integrated New York City crime records with weather, mobility, POI, public service complaints, and police activity data to jointly model cross-type, spatial, and temporal correlations [10]. Multimodal data fusion has also been explored at the feature level. One study combined Chicago crime records with socio-economic indicators, weather conditions, and street-view image features within a deep neural network (DNN)-based fusion architecture, achieving superior performance compared with traditional machine learning approaches [12].

The importance of mobility in understanding crime dynamics has also been highlighted by studies conducted during the COVID-19 pandemic. Yang et al. reported substantial changes in crime patterns across Chicago, including significant declines in theft, fraud, assault, and robbery, together with a slight increase in burglary, corresponding to mobility restrictions imposed during lockdown periods. However, their work focused on descriptive spatial-temporal analysis rather than predictive modelling [19]. Collectively, these studies demonstrate that incorporating mobility and other contextual urban data can substantially improve predictive performance. Nevertheless, none of the reviewed studies integrates mobility, weather, and socio-economic information specifically for arrest prediction.

2.4 Traditional Machine Learning and Crime-Type-Specific Studies

Several studies have applied classical machine learning techniques without incorporating deep spatiotemporal architectures. A Vancouver-based study employed k-Nearest Neighbour (k-NN) and AdaBoost algorithms for multi-class crime-type classification using temporal, geographic, and historical features. Although the proposed models demonstrated reasonable performance, they achieved comparatively lower accuracy than more advanced approaches and did not incorporate socio-economic, weather, or mobility-related variables [4]. Crime-type-specific temporal modelling has also received considerable attention. Mei

and Li compared the predictability of commercial, pedestrian, and residential robbery using Long Short-Term Memory (LSTM) networks across Atlanta and Baltimore, reporting that commercial and residential robberies were more predictable than pedestrian robberies [14]. Similarly, Ghazvini et al. proposed an Enhanced Nonlinear Autoregressive Network with Exogenous Inputs (eNARX) incorporating a novel activation function to predict serial crime characteristics, including crime location, occurrence time, and offender attributes, although the study was conducted using a relatively limited dataset [15]. Furthermore, a systematic review by Jenga et al., covering 68 crime prediction studies published between 2010 and 2022, concluded that neural networks, ensemble learning methods, and deep learning architectures have become the dominant approaches in crime prediction research. The review also highlighted persistent challenges related to data quality, class imbalance, and inconsistent evaluation methodologies, emphasizing the need for more rigorous comparative evaluations across multiple predictive models [3].

2.5 Theoretical Foundations

Much of the empirical research reviewed in this chapter is underpinned by Cohen and Felson’s Routine Activity Theory (RAT), which conceptualises crime as the convergence of a motivated offender, a suitable target, and the absence of a capable guardian in space and time [1]. Although RAT is a conceptual rather than predictive theory, it provides a strong theoretical foundation for incorporating mobility and environmental context into data-driven crime prediction models, as these variables serve as proxies for guardianship and target–offender convergence.

Across the reviewed literature, three key research gaps emerge. First, the majority of spatiotemporal deep learning studies, including the closely related work of Han et al. and Fan et al., focus on crime occurrence prediction rather than arrest outcomes, leaving the applicability of these architectures to arrest prediction largely unexplored [9, 5]. Second, studies that specifically address arrest prediction, such as that of Sankul et al., rely exclusively on traditional machine learning techniques without benchmarking their performance against deep learning or spatiotemporal architectures [6]. Third, although several studies have demonstrated the value of individual contextual data sources, including mobility [13, 17], socio-economic and multimodal information [12], and weather and Point-of-Interest (POI) data [10], no existing study integrates weather, public transportation mobility, and socio-economic indicators within a comprehensive machine learning approach that systematically compares both machine learning and deep learning models for arrest prediction.

This major research project addresses these research gaps by integrating Chicago crime records with weather observations, public transportation mobility data, and socio-economic indicators. Furthermore, it systematically evaluates the performance of a broad range of machine learning and deep learning algorithms for arrest prediction, extending the spatiotemporal modelling principles established by Han et al. and Fan et al. from crime occurrence prediction to arrest outcome prediction [9, 5].

3. Data Description and Exploratory Data Analysis

3.1 Dataset Description

This study integrates five publicly available datasets to develop a comprehensive framework for arrest prediction in the City of Chicago: crime records, socio-economic indicators, weather observations, mobility data, and community area boundaries. Each source contributes a distinct dimension – crime-level, socio-economic, environmental, or mobility-related – that may plausibly influence whether a reported crime results in an arrest. The preprocessing, integration, and feature engineering procedures used to combine these sources are described in Sections 3.2 and 3.3; this section introduces each dataset and its role in the study.

Chicago Crime Dataset. The primary dataset used in this research is the Chicago Crime Dataset, obtained from the City of Chicago Data Portal [20]. It contains reported crime incidents from 2001 to 2023 and provides the target variable used for predictive modeling, **Arrest**, together with a range of crime-level attributes: the primary crime classification (Primary Type) and a free-text description of the offense, whether the incident was domestic-related (Domestic), the police Beat and District in which it occurred, the Community Area, the corresponding FBI offense code (FBI Code), the geographic coordinates (Latitude and Longitude), and the Date and Time of occurrence. This dataset is central to the study because it provides the record-level ground truth linking incident characteristics directly to arrest outcomes.

Socio-Economic Dataset. Community-level social and economic conditions were obtained from two publicly available Chicago community-area datasets, each representing a different period of the study window. The first, covering 2008–2012, was sourced from the City of Chicago Data Portal’s Selected Socioeconomic Indicators dataset [21], which reports six indicators of public health significance – including per capita income, the unemployment rate, and the percentage of residents without a high school diploma – along with a composite hardship index, by community area. The second, a more recent 2020 community-area dataset [22], provides an expanded set of demographic, economic, and housing indicators for the same geography. Per capita income, the unemployment rate,

and the percentage of residents without a high school education were retained from each source, with the earlier snapshot representing socio-economic conditions for the earlier portion of the study period and the 2020 snapshot representing the later portion. This dataset is included because community-level economic disadvantage has been linked in prior research to variation in crime and arrest outcomes [6].

Weather Dataset. Daily weather records were obtained from the National Oceanic and Atmospheric Administration (NOAA) [23], including maximum temperature (TMAX), minimum temperature (TMIN), precipitation (PRCP), snowfall (SNOW), and average wind speed (AWND). Weather conditions are included because they shape day-to-day human activity and, in turn, opportunities for crime to occur: routine activity theory holds that a crime requires a motivated offender and a suitable target to converge in the absence of a capable guardian [1], and that convergence is itself shaped by how much people are outdoors, which weather strongly influences. Warmer, drier conditions tend to draw more people outside for longer periods, increasing the pool of potential targets and offenders in public space, while extreme heat, cold, or precipitation tends to suppress outdoor activity and, correspondingly, opportunistic street crime [10].

Mobility Dataset. Human mobility information was obtained from Chicago Transit Authority (CTA) ridership data [24], including daily bus ridership, rail boardings, total daily rides, and an indicator for weekday versus weekend service (Day Type). This dataset is included because population movement patterns capture overall urban activity levels, which prior research has linked to variation in crime exposure [13].

Chicago Community Area Boundary Dataset. The Chicago Community Area Boundary Dataset contains the geographic boundary polygons for the 77 official community areas of Chicago, obtained from the City of Chicago Data Portal [20], and includes the Community Area Number, Community Area Name, and the associated polygon boundaries. This dataset underpins the spatial and socio-economic linkage used elsewhere in the study: it allows a community area to be assigned to any crime incident based on its coordinates, which in turn enables both spatial analysis and the joining of the community-level socio-economic indicators described above.

3.2 Data Preprocessing and Integration

Before preprocessing, the Chicago Crime Dataset was first restricted to the 2001–2023 study period, removing a small number of records dated outside this range (for example, partial-year records from 2024 onward); this reduced the raw crime file from 8,498,314 to 7,980,470 records. Several preprocessing steps were then carried out to handle missing

values, remove duplicates, encode categorical variables, and integrate the five data sources described in Section 3.1 into a single analytical dataset.

Missing value handling. Each dataset was checked for missing values, and the treatment applied depended on the variable and the mechanism most consistent with its underlying structure.

In the Chicago Crime Dataset, **Community Area** had the largest share of missing values (613,682 records, 7.22%). These were recovered using a spatial join between each incident’s coordinates and the Chicago Community Area Boundary Dataset polygons: incidents were converted to point geometries and matched to the community area whose polygon contained that point. This spatial join resolved all but 9,736 records, which lacked usable coordinates and were therefore dropped, leaving 7,970,734 crime records. A small number of **District** values (47 records) were missing; because each police Beat maps to exactly one District, these were imputed using the most frequent District associated with the record’s Beat. Missing **Latitude** and **Longitude** values (83,891 records each) were imputed using the mean coordinates of all other incidents within the same Beat, on the reasoning that incidents in the same Beat are geographically close. Finally, missing **Location Description** values (13,301 records) were filled using the most frequent Location Description recorded for the same **Primary Type**, since similar crime types tend to occur in similar settings (for example, theft at a street location, burglary at a residence).

In the Socio-Economic Dataset, the 2008–2012 source contained one row missing its Community Area Number, corresponding to a citywide (*Chicago*) summary row rather than an individual community area; this row was dropped, leaving the expected 77 community-area records for that period. The 2020 source and the Weather and Mobility Datasets contained no missing values in the fields retained for this study.

Duplicate removal. Each dataset was checked for exact duplicate records prior to integration. No duplicate records were found in the mobility dataset, and the same check was applied across the other sources; consequently, no rows were removed on this basis, and the record counts reported above reflect only the missing-value treatment described.

Data integration. The cleaned sources were combined into a single analytical dataset through a sequence of key-based merges.

The Chicago Community Area Boundary Dataset was used first, as part of the missing-value treatment above, to assign a Community Area to crime incidents with missing values via the spatial join. The Socio-Economic Dataset was then integrated in two stages. First, the 2008–2012 and 2020 community-area indicator sources were combined into a single

socio-economic table, with each source tagged by its corresponding year, producing 154 rows (77 community areas $\times$ 2 periods). Second, this combined table was merged with the crime records in two parts: crime incidents from 2001–2012 were merged with the 2012 indicators, and incidents from 2013–2023 were merged with the 2020 indicators, in both cases joined on the Community Area identifier. Only two socio-economic snapshots were used, rather than a separate value for every year, because socio-economic indicators such as per capita income, unemployment, and educational attainment change gradually at the community-area level and can reasonably be treated as approximately constant within each multi-year period; the two available snapshots were therefore used to represent the earlier and later portions of the study period respectively, rather than attempting to interpolate values for years in between. The Weather Dataset was merged next, joined with the crime-and-socio-economic table on date, so that each incident received the weather conditions recorded on the day it occurred. The Mobility Dataset was merged last, also joined on date, adding daily ridership figures to each record. Where the merge produced duplicate columns already present from an earlier join (for example, `year`, `month`, and `day_of_week`, which were present in both the crime and mobility sources), the duplicated columns were verified to be identical and the redundant copies were dropped, along with identifier columns not required for modeling (`ID`, `Date`, `service_date`).

Categorical encoding. After integration, the remaining categorical predictor variables – `Primary Type`, `Description`, `Location Description`, `FBI Code`, `temp_category`, and `day_type` – were converted to numeric form using Label Encoding, which assigns each unique category an integer code. A separate encoder was fit and stored for each of these six variables. Rather than discarding this mapping once encoding was applied, the complete set of fitted encoders was saved to disk as a single serialized object, so that any encoded value can subsequently be mapped back to its original category name (for example, recovering that a given code corresponds to `THEFT` in `Primary Type` or `STREET` in `Location Description`). This preserves the interpretability of the encoded features and allows the encoding to be applied consistently to any new data using the same category-to-code mapping established on the training data.

This sequence of missing-value handling, duplicate checking, and key-based integration combined the five cleaned sources into a single integrated dataset used for the feature engineering and analysis described in the remainder of this chapter.

3.3 Feature Engineering and Feature Selection

Feature engineering. A set of additional predictors was derived from the integrated dataset to make its temporal and weather information directly usable by the predictive models.

Several temporal variables were extracted from the crime incident’s Date/Time field: `month`, `day_of_week` (coded 0 for Monday through 6 for Sunday), and `hour`, in addition to `year`, which was already present as a raw field in the crime records. A day-of-month variable, `day`, was also retained in the integrated dataset; because the crime records did not include an explicit day-of-month field, this variable was carried over from the date field of the CTA mobility source at the point of integration and applies uniformly to all records sharing the same calendar date. In addition, the `date` field itself was converted from a timestamp to a numeric (ordinal) representation so that it could be supplied directly as a model input alongside the other temporal features.

An average daily temperature, `temp_avg`, was derived as the mean of `TMAX` and `TMIN`. Several further binary and categorical indicators were also derived to test whether they captured additional signal beyond the continuous weather and temporal fields already available: an `is_weekend` flag from `day_of_week`; `is_rain` and `is_snow` flags from `PRCP` and `SNOW`; a three-level `temp_category` (cold/mild/hot) derived from `temp_avg`; and a `high_wind` flag from `AWND`. The `day_type` field, by contrast, was not derived but obtained directly from the CTA mobility source as a native weekday/weekend service indicator. Whether these engineered indicators were ultimately retained as predictors was determined by the feature-selection process described below.

Feature selection. Feature selection was carried out to identify, from the full set of engineered and integrated variables, the subset that contributed most meaningfully to arrest prediction while removing redundant or uninformative predictors. Prior to this step, the non-predictive `community_name` field was excluded, since it duplicates the information already captured numerically by `community_area` and is a free-text label rather than a usable model input; this left 34 candidate predictor variables to evaluate.

To avoid basing feature selection on data that would later be used for model evaluation, the integrated dataset was first partitioned into training (80%, 6,376,526 records) and test (20%, 1,594,132 records) sets using a stratified 80:20 split on `Arrest`, consistent with the partitioning strategy used for model development in Chapter 4. All subsequent feature-selection steps were then restricted to the training partition only, and the test set was not accessed again until final model evaluation. Because the training partition itself still contains over 6.3 million records, computing feature importance across all of it directly

was computationally impractical. A random sample of 200,000 records (approximately 3.1% of the training partition, drawn with a fixed random seed for reproducibility) was therefore used for this analysis; the sample’s arrest rate (25.70%) closely matched that of the full training partition (25.80%), indicating that the class balance of the training data was well preserved at this sample size.

A Random Forest classifier (100 trees) was trained on this training-only sample, and feature importance scores were extracted from the model’s built-in Gini-based `feature_importances_` attribute. Random Forest was chosen for this step because it natively handles a mix of categorical and continuous features without extensive preprocessing, is robust to multicollinearity among predictors, and provides a computationally efficient importance ranking. As an additional check, permutation importance was also computed on a held-out slice of the same training-only sample, which broadly corroborated the ordering obtained from the Random Forest’s built-in importance scores.

To validate this ranking and to examine each feature’s contribution at the instance level, SHapley Additive exPlanations (SHAP) were computed using `TreeExplainer` on a further sub-sample of 100 records drawn from the 200,000-record training-only sample, producing summary plots and dependence plots for key variables. In particular, the dependence plots for `bus`, `rail_boardings`, and `total_rides` confirmed a consistent relationship between mobility levels and the model’s predicted arrest probability, supporting the inclusion of the mobility variables identified as informative by the Random Forest ranking.

Based on the combined results of the Random Forest importance ranking and the SHAP analysis, seven variables were identified as having near-zero importance and were removed from the predictor set: `temp_category`, `is_rain`, `day_type`, `SNOW`, `is_weekend`, `high_wind`, and `is_snow`. These were consistently the seven lowest-ranked variables in the training-only importance analysis. For five of these, removal was further justified by redundancy with a retained variable that already captured the same information in continuous or otherwise more granular form: `is_rain` and `temp_category` were redundant with the continuous `PRCP` and `temp_avg` fields, respectively; `is_weekend` was redundant with the retained `day_of_week` field; and `high_wind` was redundant with the continuous `AWND` field. The remaining two, `SNOW` and its derived `is_snow` flag, were removed together rather than one in favour of the other: snowfall was rare in the dataset, and both the raw measurement and its binary indicator showed near-zero importance in the training-only sample. Removing them reduced the feature space from 34 candidate predictors to 27 independent predictor variables, together with the binary target variable `Arrest`. Because column selection was decided using the training partition only, this reduced set of 27 predictors was then applied to the full integrated dataset – including both the training and

test partitions – to produce the final modeling dataset used throughout the remainder of this study; the row-level train/test partition itself was preserved unchanged from the split performed at the start of this section, ensuring consistency with the model-training pipeline described in Chapter 4.

Methodological note. Feature selection in this study was performed using the training partition only. The train/test split was established before any feature-importance analysis was carried out, the 200,000-record sample used for the Random Forest and SHAP analyses was drawn exclusively from the training data, and the test set was not used or inspected at any point during this process. This avoids the test-set leakage that would result from selecting predictors using data later held out for evaluation, and ensures that the reported model performance in Chapter 5 reflects genuinely unseen data.

3.4 Final Modeling Dataset and Variables

Sections 3.1 through 3.3 described the individual data sources, the preprocessing and integration procedures used to combine them, and the feature engineering and feature-selection steps applied to arrive at a final set of predictors. This section defines the target and predictor variables that make up the final modeling dataset used throughout the remainder of this study.

Target variable. The target variable is **Arrest**, a binary indicator obtained directly from the Chicago Crime Dataset (Section 3.1), coded as 1 when an arrest was made following a reported incident and 0 otherwise.

Predictor variables. Following the feature engineering and feature-selection procedures described in Section 3.3, the final modeling dataset comprises 27 predictor variables spanning six categories: crime-related, spatial, temporal, socio-economic, weather, and mobility. Throughout this study, these are referred to as *predictor variables* rather than *independent variables*, since the predictors are not necessarily statistically independent of one another – the correlation analysis in Section 3.5 (Figure 3.6), for example, shows meaningful associations among several of the mobility variables. Table 3.1 summarizes these six categories and their representative variables. The source, definition, derivation, and preprocessing of each variable are described in full in Sections 3.1–3.3.

Final dataset. After preprocessing, integration, feature engineering, and feature selection, the final modeling dataset contains 7,970,658 records and 28 fields (27 predictor variables plus **Arrest**), spans the 2001–2023 study period, and contains no remaining missing values. The dataset exhibits a class distribution of 25.8% arrest and 74.2% non-

Table 3.1: Predictor Variable Categories

Category	Representative Variables
Crime-related	Primary Type, Description, Location Description, Domestic, FBI Code, Beat, District, Community Area
Spatial	Latitude, Longitude
Temporal	Year, Month, Day, Day of Week, Hour
Socio-economic	Per Capita Income, Unemployment Rate, No High School Percentage
Weather	TMAX, TMIN, PRCP, AWND, Average Temperature
Mobility	Bus Ridership, Rail Ridership, Total Ridership

arrest, confirming the class imbalance that motivates the resampling strategies evaluated in Chapter 4.

3.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted on the final integrated dataset (Section 3.4) to identify patterns and associations among crime incidents, socio-economic conditions, weather factors, and mobility levels, prior to any predictive modeling. The associations described in this section are descriptive: they characterize how arrest rates and crime volumes vary across time, place, and context, but they do not by themselves establish that a given variable improves predictive performance – that question is addressed empirically in Chapter 5 through the feature-importance and model-comparison results. To keep the analysis focused, this section presents the seven figures most directly relevant to the final dataset; the remaining supporting figures are provided in Appendix A.

Temporal patterns. Figure 3.1 presents annual crime counts and arrest rates in Chicago from 2001 to 2023. Reported crime declined substantially over the study period relative to the early 2000s, while the arrest rate remained comparatively stable for much of the period before declining in the later years, with some fluctuation throughout.

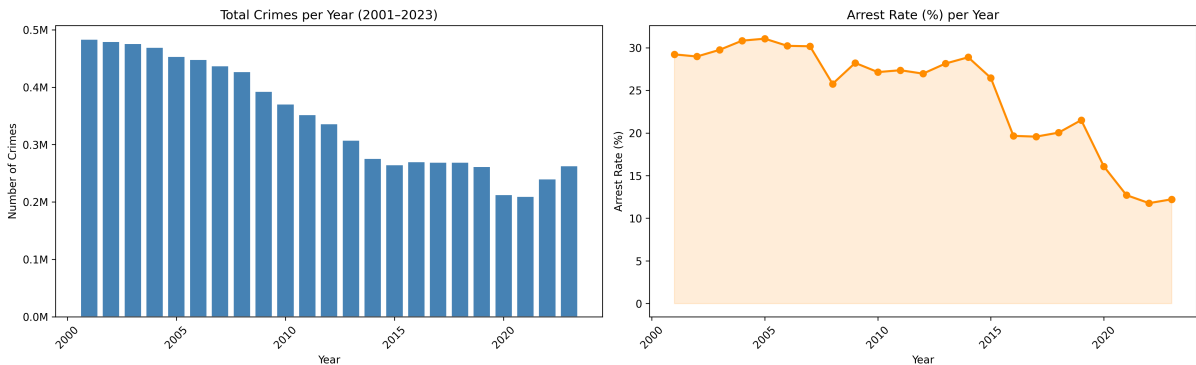


Figure 3.1: Crimes and Arrest Rate per Year

Complementary monthly, hourly, and day-of-week breakdowns, which show seasonal, daily, and weekly variation consistent with known routine-activity patterns, are provided in Appendix A (Figures A.1–A.4). These breakdowns show that both crime volume and arrest rate vary by time of day and day of week, with crime volume lowest in the early morning hours and arrest rates highest around midday.

Crime type and arrest-rate patterns. A small number of crime categories account for a large share of total incidents, led by theft, battery, criminal damage, narcotics offenses, and assault (Appendix A, Figure A.2). Figure 3.2 shows that arrest rates vary substantially across these categories: offenses such as narcotics and prostitution, which typically involve in-progress apprehension or proactive enforcement, show much higher arrest rates than offenses such as theft, burglary, or motor vehicle theft, which are more often reported after the fact.

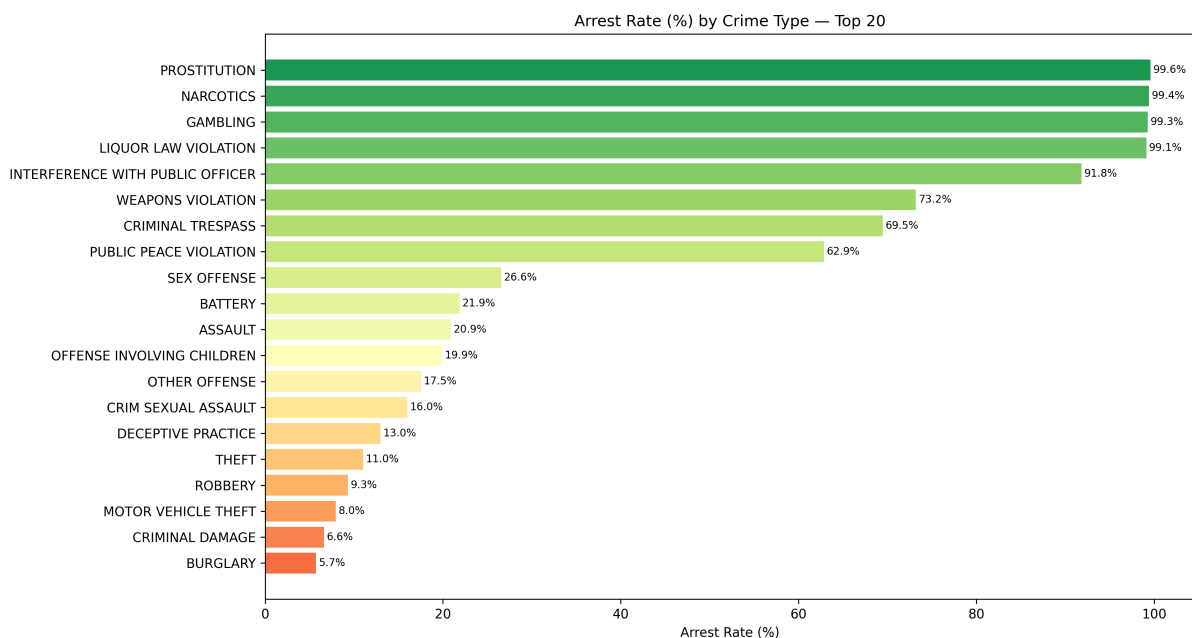


Figure 3.2: Arrest Rate by Crime Type

Spatial distribution. Figure 3.3 shows that crime is unevenly distributed across the city, with a subset of community areas – including Austin, Near North Side, and South Shore – consistently accounting for a disproportionate share of incidents. This spatial concentration is consistent with the inclusion of geographic identifiers (Community Area, Beat, District, Latitude, and Longitude) among the predictor variables described in Section 3.4.

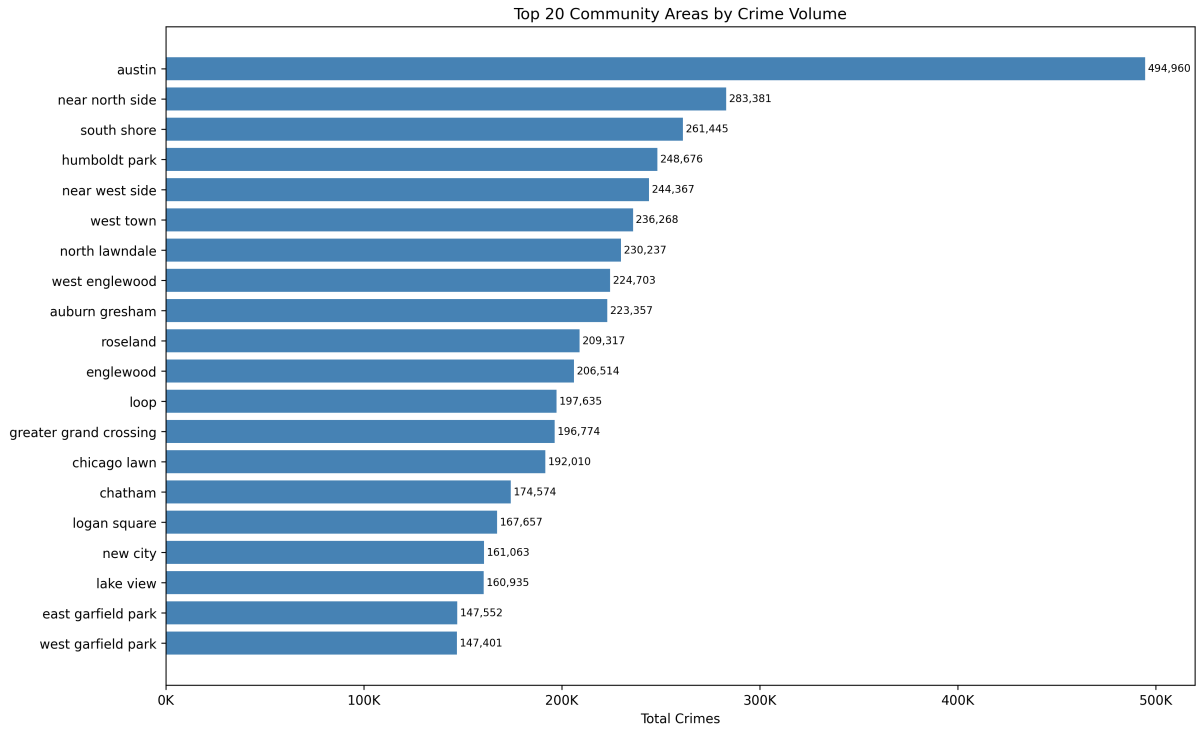


Figure 3.3: Top Community Areas by Crime Volume

Socio-economic patterns. Figure 3.4 shows that community-level socio-economic conditions – unemployment rate, educational attainment, and per capita income – are associated with variation in total crime counts across community areas, with unemployment rate showing the strongest association of the three ($r = 0.24$). A parallel breakdown of arrest rates by income and unemployment quintile, provided in Appendix A (Figure A.5), shows that arrest rates tend to be higher in lower-income and higher-unemployment community areas. These patterns are broadly consistent with prior research linking socio-economic disadvantage to variation in crime and arrest outcomes [6], though the associations observed here are moderate in strength and do not on their own indicate the predictive value of these variables.

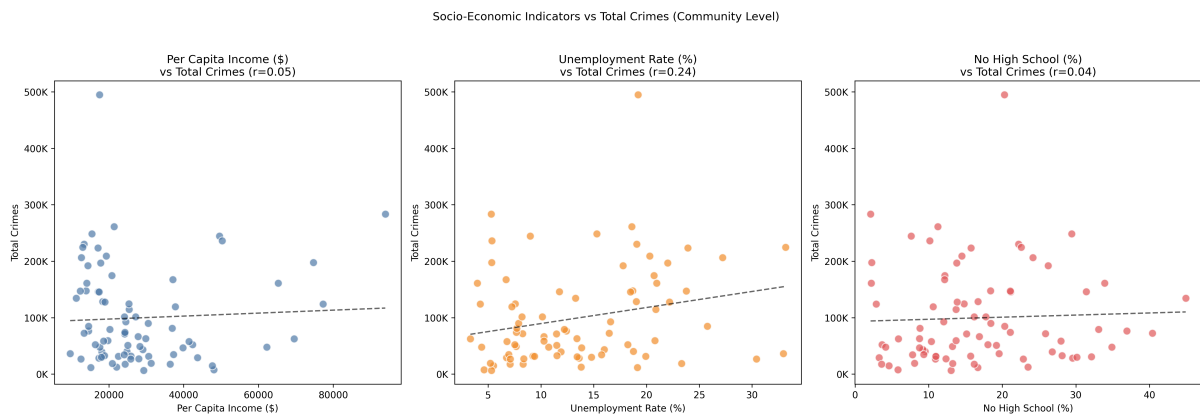


Figure 3.4: Socio-Economic Factors vs. Total Crimes

Weather patterns. Figure 3.5 shows that both crime volume and arrest rate vary by temperature category, with crime volume noticeably higher during mild and hot conditions than during cold conditions, while the arrest rate is highest during cold conditions and lowest during hot conditions. This pattern is broadly consistent with the routine-activity explanation discussed in Section 3.1, in which warmer conditions are associated with greater outdoor activity and, correspondingly, more opportunities for crime to occur.

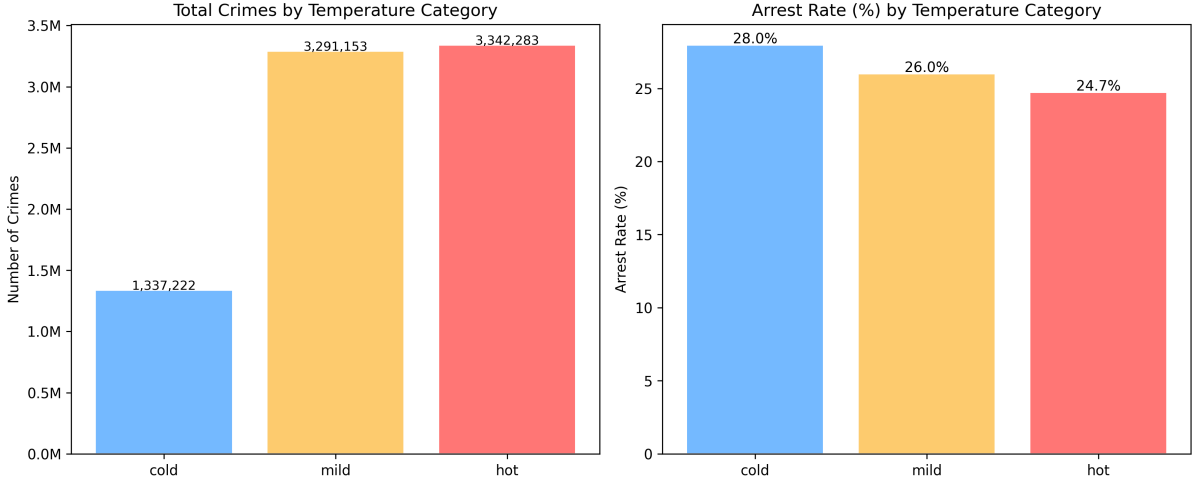


Figure 3.5: Crimes and Arrest Rate by Temperature Category

Mobility patterns. An analysis of annual crime counts against public transit ridership, provided in Appendix A (Figure A.6), shows that both series follow a broadly similar downward trajectory over the study period, with a sharp joint decline in 2020. A comparison of pre-COVID-19, COVID-19, and post-COVID-19 crime and mobility patterns, also included in Appendix A (Figure A.7), shows that the rate of crime per 1,000 transit rides was in fact highest during the COVID-19 period itself, despite the substantial decline in ridership, before partially receding in the post-COVID-19 period. This indicates that the relationship between mobility and crime volume is not a simple one-to-one association and may reflect other factors that changed concurrently during the pandemic period.

Correlation analysis. Figure 3.6 presents the correlation matrix of daily crime counts and the three mobility variables. Bus ridership and total ridership show a moderate positive association with daily crime counts ($r = 0.52$ and $r = 0.40$, respectively), while rail boardings show a weaker association ($r = 0.22$). The three mobility variables are themselves highly correlated with one another ($r \geq 0.89$), indicating a degree of redundancy among them; this redundancy is examined further through the Random Forest and SHAP feature-importance results reported in Section 3.3 and Chapter 5.

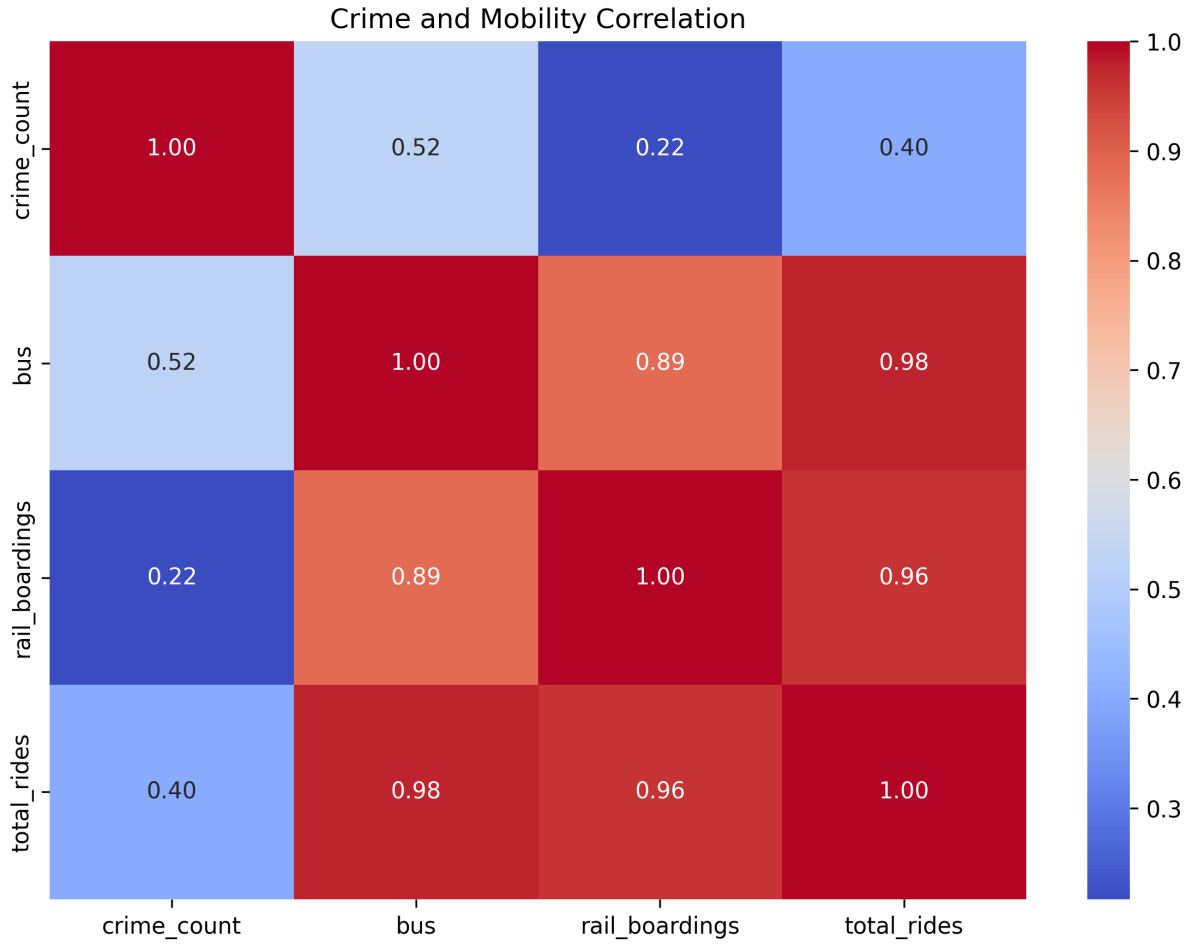


Figure 3.6: Correlation Heatmap

Taken together, the exploratory analysis shows that arrest outcomes and crime volume are associated with time, location, offense type, socio-economic context, weather conditions, and mobility levels, with no single factor dominating. These associations motivate the multi-source modeling approach adopted in this study, though the extent to which each category of variable actually contributes to predictive performance is established empirically through the feature-importance analysis in Section 3.3 and the model-comparison results in Chapter 5.

4. Methodology

4.1 Study Aim and Experimental Overview

The aim of this study is to predict whether a reported crime incident in the City of Chicago results in an arrest, using the final modeling dataset described in Chapter 3. To this end, a range of machine learning and deep learning models are trained and evaluated on the original class distribution as well as on random under-sampled and random over-sampled versions of the training data, in order to compare their effect on arrest-prediction performance. The remainder of this chapter describes the experimental design (Section 4.2), the models evaluated and how they were tuned (Section 4.3), and the metrics used to assess predictive performance (Section 4.4), with the goal of identifying the model and sampling strategy that together provide the strongest arrest-prediction performance.

4.2 Experimental Design

This section describes the modeling pipeline in the order it was executed: an 80:20 train/test split, followed by three class-balancing scenarios applied to the training data only, followed by model training and, finally, evaluation on the untouched test set. The target and final 27-predictor set used throughout this pipeline are as defined in Section 3.4.

Dataset partitioning. The final modeling dataset (Section 3.4) was divided into training and testing subsets using an 80:20 split, performed once and reused identically across every experimental scenario described below. A stratified split on **Arrest** was used to preserve the original 74.2%/25.8% class distribution in both subsets: this produced a training partition of 6,376,526 records and a test partition of 1,594,132 records, with both subsets matching the full dataset’s class balance to within 0.01 percentage points. The test partition was set aside at this point and was not accessed again – for sampling, scaling, or any other purpose – until the final evaluation step for each model.

Class-balancing scenarios. Three experimental scenarios were defined to evaluate the effect of class balancing on arrest-prediction performance, each applied to the training partition only:

1. **Original imbalanced training data.** Models were trained directly on the training partition (6,376,526 records: 4,731,082 non-arrest, 1,645,444 arrest), without any resampling.
2. **Random under-sampling.** `RandomUnderSampler` was applied to the training partition to reduce the majority (non-arrest) class to match the minority (arrest) class, producing a balanced training set of 3,290,888 records (1,645,444 per class).
3. **Random over-sampling.** `RandomOverSampler` was applied to the training partition to increase the minority (arrest) class to match the majority (non-arrest) class, producing a balanced training set of 9,462,164 records (4,731,082 per class).

In every scenario, resampling was applied strictly to the training partition produced by the split described above. The test partition retained its natural, imbalanced class distribution (74.2% non-arrest, 25.8% arrest) throughout, and every model – regardless of which training scenario produced it – was evaluated against this same untouched, unmodified test set. This ensures that the reported performance metrics in Chapter 5 reflect how each model would perform on realistically imbalanced, previously unseen data, rather than on an artificially balanced test distribution.

Where required by the model (linear and neural network architectures), feature scaling was fit using `StandardScaler` on the training data only, in each scenario, and the same fitted scaler was then applied to transform the test set – following the same train-only fitting principle used for the resampling and feature-selection steps described in Chapter 3.

4.3 Models and Hyperparameter Tuning

This section describes the machine learning and deep learning models evaluated in this study, the rationale for including each model family, and the hyperparameter configurations that produced the best performance during the experiments.

Model families. Fourteen algorithms were evaluated, spanning four complementary learning paradigms, so that the comparison in Chapter 5 reveals not only which individual model performs best but also which broad modeling approach is most suitable for arrest prediction on this dataset.

- **Linear baselines:** Logistic Regression and Linear SVM (evaluated both as a standard `LinearSVC` and as an SGD-trained linear classifier with hinge loss). These provide interpretable linear reference points against which more complex, non-linear methods can be evaluated.

- **Tree-based models:** Decision Tree, Random Forest, and Extra Trees. These represent tree-based models of increasing ensemble complexity and allow the predictive benefits of bagging and randomized splitting to be assessed.
- **Boosting models:** Gradient Boosting, AdaBoost, XGBoost, LightGBM, and CatBoost. Boosting-based ensembles have consistently achieved strong performance on structured, tabular classification problems similar to the arrest-prediction task addressed here [3].
- **Neural / deep-learning models:** Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Temporal Convolutional Network (TCN). The MLP evaluates whether a feed-forward neural network can capture non-linear interactions among the heterogeneous crime, socio-economic, weather, and mobility features. Because the dataset also includes temporal information such as year, month, day, and hour, LSTM and GRU architectures were evaluated for their ability to model temporal patterns, consistent with their use in previous crime-prediction research [9, 5]. The TCN was additionally included as a convolution-based alternative to recurrent architectures.

Comparing linear, tree-based, boosting, and neural approaches under a consistent experimental setup enables the study to determine whether the additional complexity of ensemble and neural methods provides a meaningful predictive advantage over simpler baseline models.

Hyperparameter configuration. During model development, different hyperparameter configurations were experimentally evaluated for the individual algorithms. The configurations that provided the strongest performance were retained for the final experiments. Tables 4.1 and 4.2 summarize the selected hyperparameter settings for the classical machine learning models and neural network architectures, respectively. Once the best-performing configuration for each model had been identified, the same configuration was maintained across the class-balancing scenarios described in Section 4.2. This ensured that comparisons across sampling strategies were not influenced by changes in model configuration.

Table 4.1: Hyperparameter Configuration – Classical Machine Learning Models

Model	Selected Hyperparameters
Logistic Regression	<code>max_iter=1000</code> ; remaining settings at scikit-learn defaults (L_2 penalty, lbfgs solver)
Linear SVM	<code>LinearSVC(max_iter=5000)</code> , wrapped in <code>CalibratedClassifierCV(method='sigmoid', cv=5)</code> to obtain probability estimates
SGD Linear SVM	<code>SGDClassifier(loss='hinge', penalty='l2', alpha=0.0001, max_iter=1000, tol=1e-3)</code> , wrapped in <code>CalibratedClassifierCV(cv=5)</code>
Decision Tree	scikit-learn defaults (no depth or split constraints imposed)
Random Forest	<code>n_estimators=100</code> , <code>criterion='gini'</code> , <code>max_depth=None</code> , <code>max_features='sqrt'</code> , <code>bootstrap=True</code>
Extra Trees	Same configuration as Random Forest, with <code>bootstrap=False</code>
Gradient Boosting	<code>n_estimators=100</code> , <code>learning_rate=0.1</code> , <code>max_depth=3</code> , <code>subsample=1.0</code>
AdaBoost	<code>n_estimators=100</code> , <code>learning_rate=1.0</code>
XGBoost	<code>n_estimators=100</code> , <code>learning_rate=0.1</code> , <code>max_depth=6</code> , <code>subsample=0.8</code> , <code>colsample_bytree=0.8</code>
LightGBM	<code>n_estimators=100</code> , <code>learning_rate=0.1</code> ; remaining settings at library defaults
CatBoost	<code>iterations=100</code> , <code>learning_rate=0.1</code> ; remaining settings at library defaults

Table 4.2: Hyperparameter Configuration – Neural Network Architectures

Model	Selected Hyperparameters
MLP	One hidden layer of 100 units (ReLU activation), <code>solver='adam'</code> , <code>alpha=0.0001</code> , <code>learning_rate_init=0.001</code> , <code>max_iter=100</code> , with <code>early_stopping=True</code> (10% internal validation split, <code>n_iter_no_change=10</code>)
LSTM	Single LSTM layer (64 units) → Dropout (0.3) → Dense (32, ReLU) → Dense (1, sigmoid); <code>adam</code> optimizer, binary cross-entropy loss, batch size 1024, up to 20 epochs
GRU	Single GRU layer (64 units) → Dropout (0.3) → Dense (32, ReLU) → Dense (1, sigmoid); same optimizer, loss, batch size, and epoch budget as LSTM
TCN	Temporal Convolutional layer (64 filters, kernel size 3, dilations [1, 2, 4, 8], dropout 0.3) → Dense (32, ReLU) → Dropout (0.3) → Dense (1, sigmoid); same optimizer, loss, batch size, and epoch budget as LSTM

For the neural network architectures, validation-based early stopping was used to reduce overfitting during training. For the MLP, a 10% internal validation split was used with `n_iter_no_change=10`. For the LSTM, GRU, and TCN models, 10% of the training data was similarly reserved for internal validation, and validation loss was monitored with a patience of three epochs. The model weights corresponding to the best validation performance were restored before final evaluation.

4.4 Evaluation Metrics

This section defines the metrics used to assess the predictive performance of every model evaluated in this study. These definitions apply uniformly across all fourteen models and all three class-balancing scenarios described in Sections 4.2 and 4.3.

Confusion matrix. The outcomes of binary classification are summarized using a confusion matrix (Table 4.3), which forms the basis for all metrics defined below. A *True Positive* (TP) occurs when the model correctly predicts that an incident results in an arrest; a *True Negative* (TN) occurs when the model correctly predicts that an incident does not result in an arrest; a *False Positive* (FP) occurs when the model predicts an arrest that did not occur; and a *False Negative* (FN) occurs when the model fails to predict an arrest that did occur.

Table 4.3: Confusion Matrix

	Predicted: Arrest (1)	Predicted: No Arrest (0)
Actual: Arrest (1)	True Positive (TP)	False Negative (FN)
Actual: No Arrest (0)	False Positive (FP)	True Negative (TN)

Accuracy. Accuracy measures the overall proportion of correctly classified incidents:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.1)$$

Accuracy is reported for completeness but is not the primary metric of interest in this study, since a classifier can achieve high accuracy simply by predicting the majority (non-arrest) class most of the time.

Precision. Precision measures the proportion of predicted arrests that were correct:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4.2)$$

Higher precision indicates fewer false-positive arrest predictions.

Recall. Recall (sensitivity, or the true positive rate) measures the proportion of actual arrests that the model correctly identifies:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4.3)$$

F1-score. The F1-score is the harmonic mean of Precision and Recall, providing a single measure that balances false positives and false negatives:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.4)$$

ROC-AUC. The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate against the False Positive Rate as the classification threshold is varied. The True Positive Rate is equivalent to Recall (Equation 4.3), and the False Positive Rate is defined as:

$$\text{FPR} = \frac{FP}{FP + TN} \quad (4.5)$$

The Area Under the ROC Curve is given by:

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(f) df, \quad f = \text{FPR} \quad (4.6)$$

where the integral is evaluated over the ROC curve traced out across all classification

thresholds. ROC-AUC summarizes a classifier’s ability to distinguish between arrest and non-arrest cases independently of any single threshold, with a value of 1.0 indicating perfect separation and 0.5 indicating performance equivalent to random guessing.

PR-AUC. The Precision–Recall (PR) curve plots Precision (Equation 4.2) against Recall (Equation 4.3) as the classification threshold is varied. The Area Under the Precision–Recall Curve is given by:

$$\text{PR-AUC} = \int_0^1 \text{Precision}(r) \, dr, \quad r = \text{Recall} \quad (4.7)$$

where the integral is evaluated over the PR curve traced out across all classification thresholds. PR-AUC summarizes the precision–recall trade-off into a single threshold-independent measure, focused specifically on performance with respect to the positive (arrest) class.

Metric Selection for Class-Imbalanced Data. Because the final dataset is imbalanced (74.2% non-arrest, 25.8% arrest, as established in Section 3.4), Accuracy alone can be misleading: a model that predicts the majority class for nearly every incident can still achieve a high accuracy score while failing to identify most actual arrest cases. Recall and F1-score are therefore emphasized because they directly measure how well a model identifies the minority (arrest) class rather than rewarding majority-class prediction. ROC-AUC is used as a threshold-independent measure of overall discriminative ability, allowing models to be compared without committing to a single decision threshold. PR-AUC is given particular weight in this study, since – unlike ROC-AUC, which can appear favorable even when a classifier performs poorly on the minority class in a heavily imbalanced setting – PR-AUC is calculated with respect to the positive class directly, making it a more sensitive and realistic measure of a model’s ability to correctly identify arrest cases specifically, which is the outcome of primary practical interest in this study.

5. Results and Discussion

5.1 Results by Sampling Strategy

This section reports the test-set performance of the fourteen models described in Section 4.3 under each of the three class-balancing scenarios defined in Section 4.2: the original imbalanced training data, random under-sampling, and random over-sampling. In every case, models were trained on the corresponding training partition and evaluated on the same untouched, naturally imbalanced test set (74.2% non-arrest, 25.8% arrest), so that the reported metrics are directly comparable across scenarios. Models are ordered by F1-score within each table, since F1-score balances Precision and Recall and is the most representative single metric for this imbalanced classification task (Section 4.4).

Original imbalanced dataset. Table 5.1 reports model performance when trained directly on the original, imbalanced training partition. Random Forest, LightGBM, XGBoost, and Extra Trees produced the strongest results overall, each combining high Accuracy (above 0.88) with the highest ROC-AUC and PR-AUC values obtained under this scenario (ROC-AUC > 0.90 , PR-AUC > 0.84). Despite these otherwise strong results, every model showed a pronounced gap between Precision and Recall for the arrest class: even the top-performing models recovered only 60–65% of actual arrest cases (Recall), while linear models such as Logistic Regression and Linear SVM identified fewer than half. This pattern indicates that, although Accuracy remains high across all models, training on the original class distribution biases every classifier toward the majority non-arrest class and limits their ability to correctly identify the minority arrest class – the primary motivation for evaluating the balancing strategies in the sections that follow.

Table 5.1: Model Performance on the Original Imbalanced Training Data

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Random Forest	0.8936	0.9086	0.6534	0.7602	0.9116	0.8549
Extra Trees	0.8889	0.8917	0.6483	0.7507	0.9047	0.8449
LightGBM	0.8900	0.9111	0.6358	0.7490	0.9109	0.8532
XGBoost	0.8896	0.9068	0.6375	0.7487	0.9112	0.8535
TCN	0.8872	0.8941	0.6387	0.7451	0.9073	0.8486
CatBoost	0.8867	0.9043	0.6272	0.7407	0.9045	0.8446
LSTM	0.8855	0.8923	0.6326	0.7403	0.9046	0.8442
GRU	0.8851	0.8900	0.6330	0.7398	0.9038	0.8431
MLP	0.8833	0.8893	0.6258	0.7347	0.9004	0.8384
Gradient Boosting	0.8817	0.9000	0.6095	0.7268	0.8976	0.8365
AdaBoost	0.8665	0.8884	0.5519	0.6808	0.8768	0.7984
Decision Tree	0.8316	0.6668	0.6940	0.6801	0.7868	0.5419
SGD Linear SVM	0.8220	0.7231	0.5024	0.5929	0.7333	0.5037
Linear SVM	0.8198	0.7228	0.4896	0.5838	0.7471	0.5261
Logistic Regression	0.8181	0.7222	0.4798	0.5766	0.7509	0.5388

Random under-sampling. Table 5.2 reports model performance after applying random under-sampling to the training partition, so that arrest and non-arrest cases were equally represented during training. Relative to the original imbalanced setting, Recall improved substantially across almost every model, most notably for the tree-based ensembles: Random Forest’s Recall rose from 0.6534 to 0.7659, and Extra Trees’ from 0.6483 to 0.7637, with comparable gains for XGBoost, LightGBM, and CatBoost. F1-score also improved for most models, though less uniformly, since the Recall gains were partly offset by a corresponding drop in Precision – Random Forest’s Precision, for example, fell from 0.9086 to 0.7140. Accuracy declined across the board relative to the original dataset, consistent with the more balanced but harder decision boundary induced by the reduced training set. ROC-AUC and PR-AUC values remained broadly similar to those obtained on the original data for the strongest models, with Random Forest, XGBoost, and LightGBM again producing the highest values (ROC-AUC > 0.91, PR-AUC > 0.85).

Table 5.2: Model Performance with Random Under-Sampling

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
TCN	0.8750	0.7950	0.6950	0.7416	0.9033	0.8424
XGBoost	0.8640	0.7313	0.7479	0.7395	0.9113	0.8527
Random Forest	0.8604	0.7140	0.7659	0.7391	0.9133	0.8545
LightGBM	0.8630	0.7284	0.7479	0.7380	0.9109	0.8520
CatBoost	0.8624	0.7318	0.7368	0.7343	0.9054	0.8440
LSTM	0.8680	0.7702	0.6960	0.7312	0.8981	0.8335
Gradient Boosting	0.8628	0.7409	0.7202	0.7304	0.8990	0.8360
GRU	0.8692	0.7835	0.6817	0.7290	0.8939	0.8271
Extra Trees	0.8533	0.6969	0.7637	0.7288	0.9070	0.8453
MLP	0.8496	0.6920	0.7515	0.7205	0.9013	0.8389
AdaBoost	0.8326	0.6721	0.6861	0.6790	0.8743	0.7949
Decision Tree	0.7698	0.5372	0.7798	0.6361	0.7731	0.4758
Logistic Regression	0.7479	0.5092	0.6403	0.5673	0.7546	0.5247
Linear SVM	0.7478	0.5090	0.6397	0.5669	0.7539	0.5244
SGD Linear SVM	0.7417	0.4995	0.6297	0.5571	0.7389	0.4913

Random over-sampling. Table 5.3 reports model performance after applying random over-sampling to the training partition, so that the minority arrest class was duplicated to match the size of the majority non-arrest class while retaining every original majority-class observation. Relative to the original imbalanced dataset, Recall again improved for nearly every model – Random Forest rose from 0.6534 to 0.6760 and XGBoost from 0.6375 to 0.7489 – while Accuracy and Precision remained closer to their original-dataset values than was the case under under-sampling. Random Forest achieved the highest F1-score (0.7617), ROC-AUC (0.9122), and PR-AUC (0.8545) of any model in this scenario, marginally exceeding its own performance on the original dataset on all three metrics simultaneously. Compared with random under-sampling (Table 5.2), over-sampling produced similar Recall and F1-score for the boosting models (XGBoost, LightGBM, CatBoost, Gradient Boosting) but generally higher Precision and Accuracy, consistent with over-sampling preserving the full set of original majority-class observations rather than discarding a portion of them.

Table 5.3: Model Performance with Random Over-Sampling

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Random Forest	0.8908	0.8723	0.6760	0.7617	0.9122	0.8545
Extra Trees	0.8890	0.8965	0.6440	0.7496	0.9044	0.8444
LSTM	0.8709	0.7708	0.7110	0.7397	0.9049	0.8441
XGBoost	0.8639	0.7305	0.7489	0.7396	0.9115	0.8530
LightGBM	0.8635	0.7303	0.7469	0.7385	0.9108	0.8520
TCN	0.8752	0.8063	0.6798	0.7376	0.8999	0.8372
CatBoost	0.8623	0.7317	0.7365	0.7341	0.9057	0.8443
GRU	0.8681	0.7725	0.6929	0.7305	0.8956	0.8303
Gradient Boosting	0.8627	0.7429	0.7157	0.7291	0.8986	0.8352
MLP	0.8530	0.7026	0.7464	0.7238	0.9026	0.8398
Decision Tree	0.8341	0.6748	0.6892	0.6819	0.7869	0.5454
AdaBoost	0.8295	0.6624	0.6919	0.6768	0.8758	0.7973
Logistic Regression	0.7478	0.5091	0.6404	0.5672	0.7547	0.5248
Linear SVM	0.7477	0.5089	0.6399	0.5669	0.7540	0.5245
SGD Linear SVM	0.7416	0.4994	0.6289	0.5567	0.7387	0.4902

5.2 Overall Model and Sampling Comparison

Section 5.1 reported model performance separately for each of the three class-balancing scenarios. This section brings those results together to compare all fourteen models and three sampling strategies directly, identify the models and metrics that consistently rank highest, and select the model/sampling combination that provides the strongest overall arrest-prediction performance.

Average performance across sampling strategies. Figure 5.1 summarizes the average value of each evaluation metric, taken across all fourteen models, for the original, under-sampled, and over-sampled training scenarios. The original imbalanced dataset produced the highest average Accuracy and Precision, consistent with the tendency of models trained on the natural class distribution to favour the majority non-arrest class. However, both balancing strategies produced a substantially higher average Recall than the original dataset, with a corresponding improvement in average F1-score. Average ROC-AUC and PR-AUC were similar across all three scenarios, indicating that class balancing altered the operating point of the classifiers – shifting the trade-off between Precision and Recall – rather than the models’ underlying ability to discriminate between arrest and non-arrest cases. Between the two balancing strategies, random over-sampling produced marginally higher average Accuracy, Precision, and F1-score than

random under-sampling, while random under-sampling produced marginally higher average Recall.

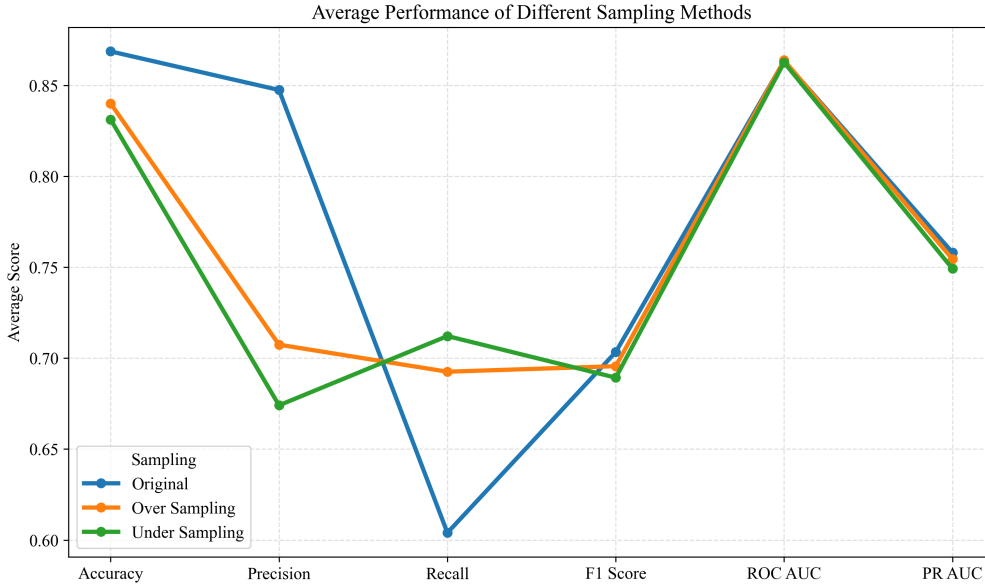


Figure 5.1: Average Performance of Different Sampling Methods

Top-performing models. Figure 5.2 ranks the ten best-performing model/sampling combinations by F1-score across all experiments reported in Section 5.1. Ensemble learning algorithms dominate the ranking: Random Forest, Extra Trees, LightGBM, XGBoost, and CatBoost occupy five of the top seven positions, with the Temporal Convolutional Network (TCN) and LSTM the only non-ensemble architectures to appear in the top ten. This ranking is consistent with the per-scenario results in Section 5.1, in which tree-based and boosting ensembles consistently outperformed linear and, to a lesser extent, deep-learning models across all three sampling conditions.

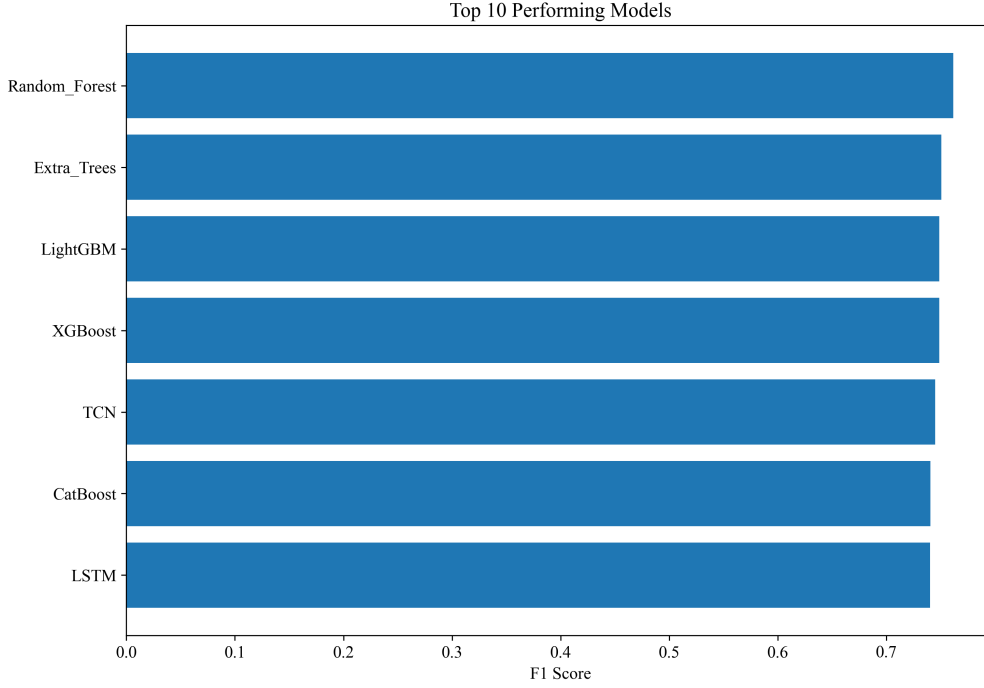


Figure 5.2: Top 10 Performing Models

F1-score comparison. Figure 5.3 compares F1-score across all fourteen models and three sampling strategies directly. Random Forest achieves the highest F1-score of any model under both the original and over-sampled scenarios (Tables 5.1 and 5.3), while XGBoost and TCN produce the highest F1-scores under under-sampling (Table 5.2). Across all three sampling strategies, boosting and tree-based ensembles (Random Forest, XGBoost, LightGBM, CatBoost, Extra Trees) occupy the upper half of the ranking, whereas the linear models (Logistic Regression, Linear SVM, SGD Linear SVM) remain at the bottom regardless of sampling strategy. This indicates that the relative ordering of models by F1-score is largely stable across sampling strategies, even though the balancing strategy materially shifts each model’s absolute Precision/Recall trade-off, as reported in Section 5.1.

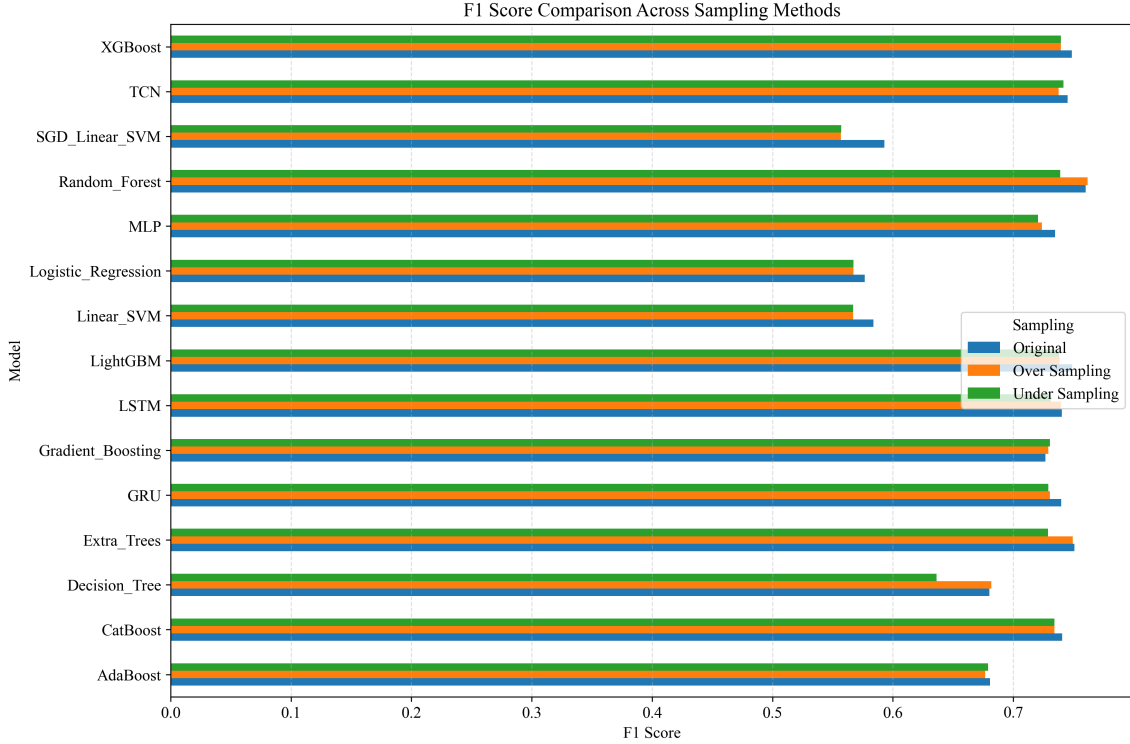


Figure 5.3: F1-Score Comparison Across Sampling Methods

F1-score is retained here as the single representative metric because it balances Precision and Recall in one figure; the corresponding Accuracy, Precision, and Recall comparisons across all models and sampling strategies follow the same overall ranking pattern and are provided for readers who wish to inspect each metric individually in Appendix B (Figures B.1–B.3).

ROC-AUC comparison. Figure 5.4 compares ROC-AUC across all evaluated models. Random Forest obtains the highest ROC-AUC overall (0.9133, under random under-sampling; Table 5.2), closely followed by XGBoost and LightGBM, with each of these three models exceeding 0.91 under every sampling strategy. Ensemble and boosting models consistently achieve ROC-AUC values above 0.90 regardless of sampling strategy, while the linear models (Logistic Regression, Linear SVM, SGD Linear SVM) remain below 0.76 in every scenario. Unlike F1-score, ROC-AUC changes only marginally across sampling strategies for any individual model, reinforcing the observation above that class balancing primarily shifts the Precision–Recall operating point rather than a model’s underlying discriminative capacity.

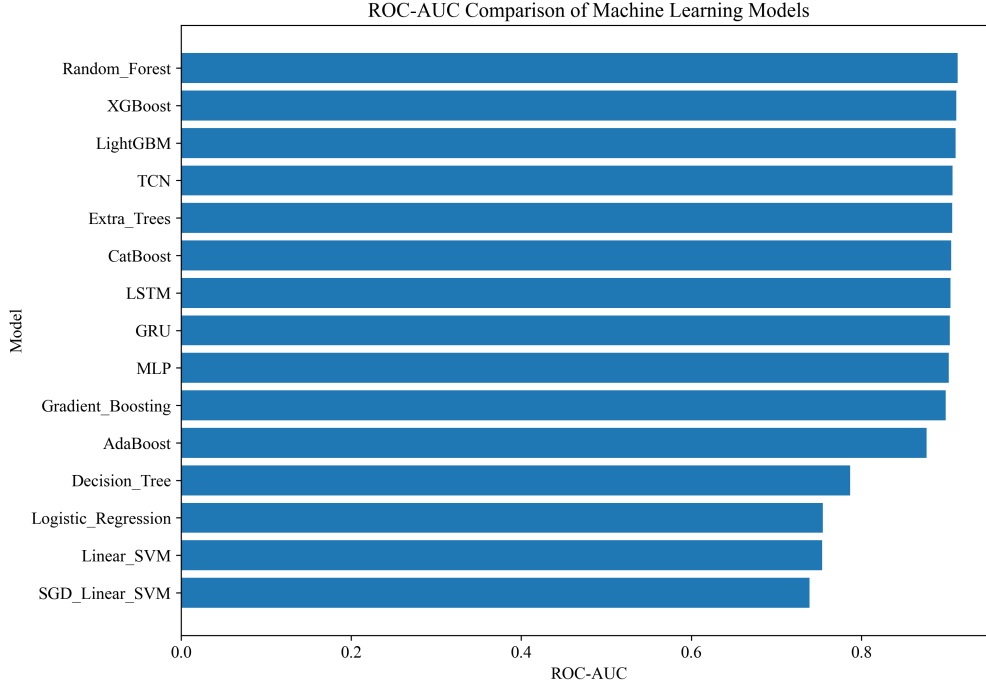


Figure 5.4: ROC-AUC Comparison of Machine Learning Models

PR-AUC comparison. Figure 5.5 compares PR-AUC across all evaluated models. Because PR-AUC is calculated with respect to the positive (arrest) class and is therefore more sensitive to minority-class performance than ROC-AUC (Section 4.4), it provides the most direct evidence of which models are best suited to arrest prediction specifically. Random Forest again achieves the highest PR-AUC of any model (0.8549, under the original imbalanced dataset; Table 5.1), with XGBoost, LightGBM, CatBoost, and Extra Trees each exceeding 0.84 across all three sampling strategies. As with ROC-AUC, PR-AUC values are similar across the original, under-sampled, and over-sampled scenarios for the top-performing models, while the linear models and Decision Tree remain well below 0.55 in every scenario.

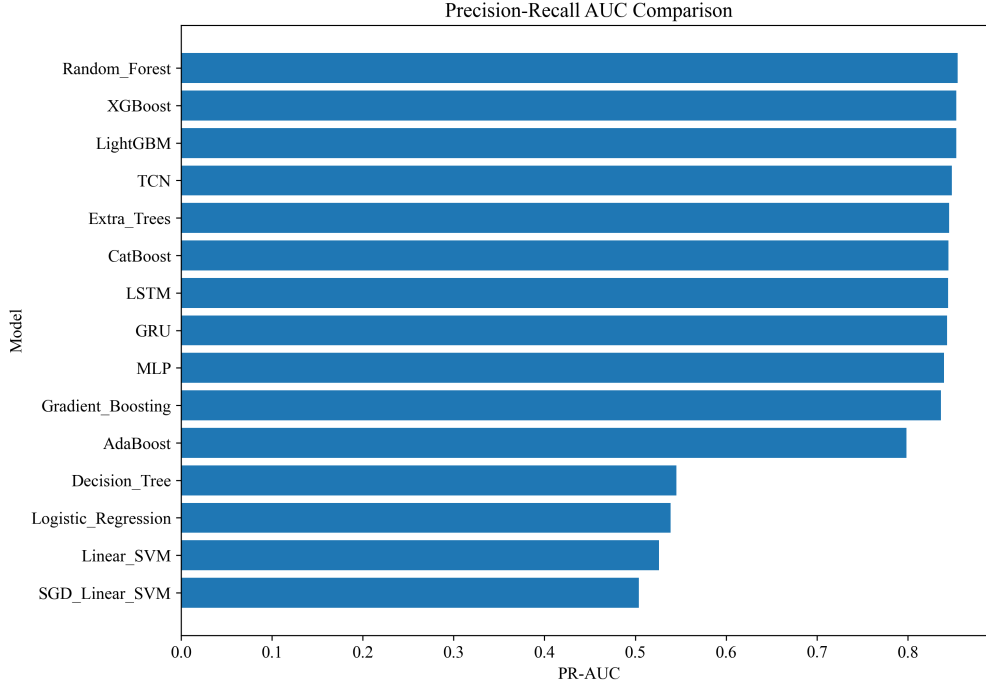


Figure 5.5: Precision–Recall Curves of the Best-Performing Models

Final model selection. Taken together, the comparisons in this section point consistently to Random Forest as the strongest individual model, and to random over-sampling as the sampling strategy that provides the best overall balance of performance. Random Forest achieves the highest or near-highest F1-score, ROC-AUC, and PR-AUC under every sampling strategy evaluated (Tables 5.1–5.3), and no other model surpasses it on more than one of these three metrics simultaneously in any scenario.

Among the three Random Forest configurations, the model trained with random over-sampling is selected as the final arrest-prediction model. It achieves the highest F1-score of the three (0.7617, versus 0.7602 on the original data and 0.7391 under under-sampling), a ROC-AUC (0.9122) and PR-AUC (0.8545) within 0.001–0.002 of the best value obtained by any Random Forest variant, and a substantially higher Recall than the original-dataset model (0.6760 versus 0.6534) without the pronounced loss of Precision observed under under-sampling (0.8723 versus 0.7140). This combination therefore offers the most favourable trade-off between correctly identifying arrest cases and avoiding false-positive arrest predictions, while preserving the full set of original training observations rather than discarding data, as random under-sampling does. On this basis, Random Forest trained on the randomly over-sampled dataset is adopted as the final model for the arrest-prediction framework developed in this study.

5.3 Discussion

This section interprets the results reported in Sections 5.1 and 5.2, and considers what they imply about the feasibility of arrest prediction, the contribution of the different predictor groups, and the role of class imbalance in shaping model behaviour.

5.3.1 Arrest Prediction Performance

The results indicate that arrest prediction is feasible, but only to a moderate degree, from the crime, socio-economic, weather, and mobility data integrated in this study. The best-performing configuration, Random Forest trained with random over-sampling, achieved a ROC-AUC of 0.9122 and a PR-AUC of 0.8545 (Section 5.2), indicating that the model can discriminate between arrest and non-arrest incidents considerably better than chance and substantially better than a naive baseline that always predicts the majority class. At the same time, its Recall (0.6760) shows that roughly one in three actual arrests were still missed at the default 0.5 threshold, and no model or sampling strategy evaluated in this study exceeded an F1-score of 0.77 (Tables 5.1–5.3). Arrest outcomes are therefore only partially explained by the variables available here; a meaningful share of the variation is likely driven by factors not captured in the dataset, such as officer discretion, real-time operational conditions, and case-specific evidence, consistent with the limitations noted below.

Across all three sampling strategies, tree-based and boosting ensembles – Random Forest, Extra Trees, XGBoost, LightGBM, and CatBoost – consistently outperformed both the linear baselines (Logistic Regression, Linear SVM, SGD Linear SVM) and the sequence-based deep learning models (LSTM, GRU, TCN) on every metric reported (Section 5.2). The gap relative to the linear models was the largest and most consistent finding in the comparison: Logistic Regression and the two SVM variants trailed the ensembles by roughly 0.15–0.20 in F1-score and 0.15 in ROC-AUC in every scenario. This is consistent with the nature of the predictor set, which mixes categorical crime and location variables with continuous weather and mobility measures and is unlikely to relate to arrest outcomes through a single linear boundary; tree-based models can partition this heterogeneous feature space into local regions and capture interactions (for example, between crime type and time of day, or between community area and socio-economic conditions) without those interactions having to be specified in advance.

The comparison with the deep learning architectures is more modest but still favours the ensembles. LSTM, GRU, and TCN achieved competitive but not superior perfor-

mance, typically ranking just behind the top ensemble models on F1-score, ROC-AUC, and PR-AUC (Section 5.2). This is plausible given the structure of the dataset: each crime incident was treated as an independent record with a set of temporal features (year, month, day, hour) rather than as part of an explicit sequence, so the sequence-modelling capacity of the LSTM and GRU architectures had comparatively little sequential structure to exploit. The tabular, mixed-type nature of the final feature set – twenty-seven variables spanning crime, spatial, temporal, socio-economic, weather, and mobility categories (Table 3.1) – is the kind of setting in which tree-based ensembles have consistently been reported to perform well in the wider literature (Section 2.4), and the results obtained here are consistent with that pattern rather than an exception to it.

5.3.2 Contribution of Predictor Groups

The exploratory analysis in Chapter 3, together with the Random Forest and SHAP-based feature analysis described in Section 3.3, points to crime-related, temporal, and spatial variables as the predictor groups most directly associated with arrest outcomes, with socio-economic, weather, and mobility variables contributing additional but comparatively weaker context. Arrest rates varied sharply by crime type, from under 10% for burglary and criminal damage to over 90% for narcotics, gambling, and liquor-law violations (Figure 3.2), a spread far larger than that observed for any other predictor group examined in this study. This is consistent with the underlying nature of these offences: narcotics, gambling, and liquor-law violations are typically detected through direct police observation or enforcement action, which results in an arrest by construction, whereas offences such as burglary and theft are usually reported after the fact, when the offender is no longer present. Crime type is therefore expected to carry substantial predictive weight, and its inclusion in the feature set is well justified by the EDA alone.

Spatial and temporal variables also showed clear structure in the exploratory analysis: crime volume was highly concentrated in a subset of community areas (Figure 3.3), and both crime counts and arrest rates varied by hour, day of week, and month (Figures 3.1, A.3, A.4, A.1). These patterns motivate the inclusion of location and time variables in the model, though the EDA on its own cannot separate how much of this variation reflects genuine differences in the likelihood of arrest as opposed to differences in reporting or enforcement patterns across areas and times of day.

The socio-economic, weather, and mobility variables showed real but comparatively weak associations in the exploratory analysis. Per-capita income, unemployment rate, and the percentage of residents without a high-school education were only weakly correlated with total crime counts at the community-area level (Pearson $r = 0.05, 0.24, \text{ and } 0.04$

respectively; Figure 3.4), and the corresponding breakdown of arrest rate by income and unemployment quintile in Figure A.5 shows only a modest gradient across quintiles rather than a sharp separation. Weather effects on crime volume were visible but modest – crime counts were noticeably lower in cold conditions than in mild or hot conditions, while the arrest rate itself varied by only a few percentage points across temperature categories (Figure 3.5) – and the correlation analysis in Section 3.3 was strong enough among the raw weather measures (e.g., TMAX and TMIN) and the derived weather flags that several of the flags were removed as redundant prior to modelling. The mobility variables were highly correlated with one another (bus, rail, and total ridership correlated at $r \geq 0.89$; Figure 3.6) and only moderately correlated with total crime volume ($r = 0.52$ for bus ridership, 0.22 for rail boardings, 0.40 for total rides), and the SHAP analysis described in Section 3.3 was used specifically to examine the individual contribution of these three ridership variables given their strong mutual redundancy.

Taken together, these findings support the inclusion of socio-economic, weather, and mobility variables as complementary contextual information, consistent with their role in the routine-activity framework discussed in Section 2.5, rather than as dominant predictors of arrest outcomes in their own right. It is important to note, however, that this study did not train separate models with and without each predictor group, so no claim is made here that adding socio-economic, weather, or mobility data produced a measurable improvement in predictive performance; the Random Forest importance ranking and SHAP analysis in Section 3.3 describe the relative contribution of variables *within* the full model, not the effect of removing a given data source. Establishing the specific performance contribution of each predictor group would require a dedicated ablation study, which is identified as a direction for future work below.

5.3.3 Class Imbalance

The comparison across the original, under-sampled, and over-sampled training scenarios in Sections 5.1 and 5.2 shows that class imbalance has a substantial and predictable effect on model behaviour. Models trained on the original, imbalanced data achieved the highest Accuracy and Precision but the lowest Recall of the three scenarios, reflecting a bias toward the majority non-arrest class: a classifier that is cautious about predicting an arrest will be right more often overall, simply because non-arrest is the more common outcome, but it will also miss a larger share of the arrests that do occur. Both random under-sampling and random over-sampling corrected this bias by exposing the models to a more balanced representation of the minority class during training, which raised average Recall and F1-score at the cost of average Accuracy and Precision (Figure 5.1).

This pattern is a direct illustration of the precision–recall trade-off rather than a genuine improvement or deterioration in the underlying models: ROC-AUC and PR-AUC, which summarize performance across all classification thresholds rather than at a single operating point, remained broadly similar across the three sampling strategies for the strongest models (Section 5.2). This indicates that balancing the training data shifted *where* each model’s decision boundary sits between the two classes – trading some Precision for Recall – rather than changing how well the model could separate arrest from non-arrest cases in principle. Whether this trade-off is desirable depends on the intended use of the model: because failing to flag a case that would in fact result in an arrest (a false negative) is arguably more costly than an occasional false-positive flag in a decision-support setting, prioritizing Recall through class balancing is appropriate for the stated goal of supporting evidence-based policing and resource allocation.

Between the two balancing strategies, random over-sampling is judged the more appropriate choice for this study, and was accordingly used for the final model selected in Section 5.2. Random under-sampling discards a large share of the majority-class observations to match the size of the minority class, which, given that non-arrest incidents make up 74.2% of the 7,970,658-record dataset, removes several million potentially informative records from training. Random over-sampling instead retains every original observation and duplicates minority-class records to achieve balance, which avoids this loss of information; this is reflected in the results, where over-sampled models achieved Recall gains comparable to the under-sampled models while retaining Accuracy and Precision closer to the original-dataset values (Section 5.1). The main limitation of over-sampling – an increased risk of overfitting to duplicated minority-class examples, and longer training time due to the larger resulting training set – was not found to outweigh this advantage for the tree-based ensembles that performed best in this study, since Random Forest’s over-sampled Precision (0.8723) remained close to its original-dataset value (0.9086) rather than collapsing toward the level observed under under-sampling (0.7140).

5.4 Limitations and Future Research

The results and interpretation presented above should be understood within the context of several limitations related to the data, features, and modelling approach used in this study. These limitations are discussed separately here, rather than as part of the main discussion, to clearly define the scope of the findings and to motivate the directions for future research proposed at the end of this section.

Single-city dataset and generalizability. The framework developed in this study

was trained and evaluated exclusively on crime incidents from a single metropolitan area, the City of Chicago, spanning 2001 to 2023 (Section 3.1). Chicago’s policing practices, reporting procedures, crime-type mix, and socio-economic and mobility patterns are specific to that city, and the relationships learned by the models – for example, the strong association between certain crime types and arrest likelihood observed in Section 5.3.2 – may not transfer directly to cities with different enforcement priorities, demographic composition, or data-collection practices. Without evaluating the framework on data from other jurisdictions, it is not possible to determine how much of the reported predictive performance reflects patterns that are general to urban crime and arrest processes, and how much reflects characteristics particular to Chicago. Generalizability across cities therefore remains untested and should not be assumed.

Reliance on historical, publicly available variables. The predictor set was constructed entirely from publicly available historical records: crime incidents, community-level socio-economic indicators, NOAA weather observations, and CTA ridership data (Section 3.1). These sources were not designed for arrest prediction and impose several structural constraints on the resulting model. The socio-economic indicators, in particular, were only available at two fixed points in time (2008–2012 and 2020) and had to be assigned to crime records using a piecewise-static join by year range (Section 3.2), meaning the model cannot capture year-to-year changes in community-level conditions within either period. More generally, because every variable in the model is drawn from historical, retrospective sources, the framework is inherently limited to whatever signal is captured in administrative and open-data records, and cannot draw on information that is not routinely collected or published in this form.

Unobserved operational factors. Several factors that plausibly influence whether a reported crime results in an arrest are not present in any of the five integrated data sources and could not be included in the model. These include officer deployment levels and patrol allocation at the time and location of the incident, real-time operational conditions such as staffing shortages or competing incidents, the strength of the physical or witness evidence available in a specific case, and investigative decisions made by individual officers or detectives. The moderate Recall and F1-score obtained by the best-performing model (Section 5.3.1) are consistent with a meaningful share of arrest outcomes being driven by exactly this kind of case-specific, operational information that lies outside the scope of the crime, spatial, temporal, socio-economic, weather, and mobility variables used here. Any application of this framework to operational decision-making should account for this gap rather than treating the model’s predictions as capturing the full set of factors that determine an arrest.

Future research directions. Building on the limitations above, several extensions to

the present framework would be a natural continuation of this work. First, incorporating explicit spatial-temporal deep learning architectures – such as the LSTM-ST-GCN and Informer-based models reviewed in Section 2.1 – rather than treating each crime incident as an independent record, could allow future work to more directly test whether modelling spatial and temporal dependencies jointly improves arrest prediction, extending the crime-occurrence-focused findings of that literature to the arrest-prediction task addressed here. Second, graph neural networks that represent community areas, police beats, or districts as nodes in a spatial graph could capture geographic spillover effects between neighbouring areas that the beat- and community-area-level features used in this study cannot represent directly. Third, applying explainable AI techniques, extending beyond the Random Forest importance and SHAP analysis already used for feature selection in Section 3.3, to the final selected model itself would allow individual arrest predictions to be interpreted and audited, which is an important consideration for any model intended to inform policing decisions. Finally, integrating real-time or near-real-time data – such as current officer deployment, live weather conditions, or same-day mobility figures, rather than the retrospective sources used in this study – would move the framework closer to a deployable decision-support tool, though this would require access to operational data sources beyond the publicly available records used here. Each of these directions extends the scope of the present study rather than falling within it, and is proposed as a basis for future work rather than a claim about what the current framework achieves.

A. Supplementary Exploratory Data Analysis Figures

This appendix presents the exploratory figures that support but are not essential to the main narrative in Chapter 3. They are retained here for completeness and to allow readers to verify the finer-grained patterns summarized in the main text.

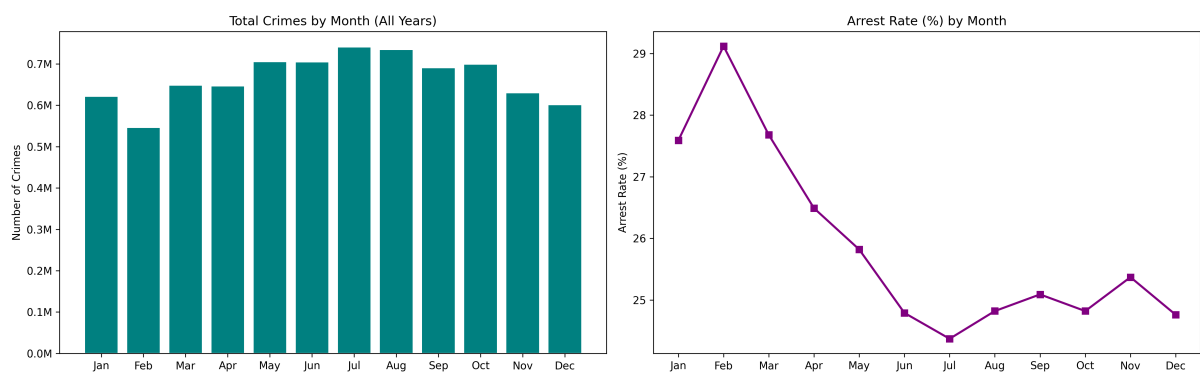


Figure A.1: Crimes and Arrest Rate by Month

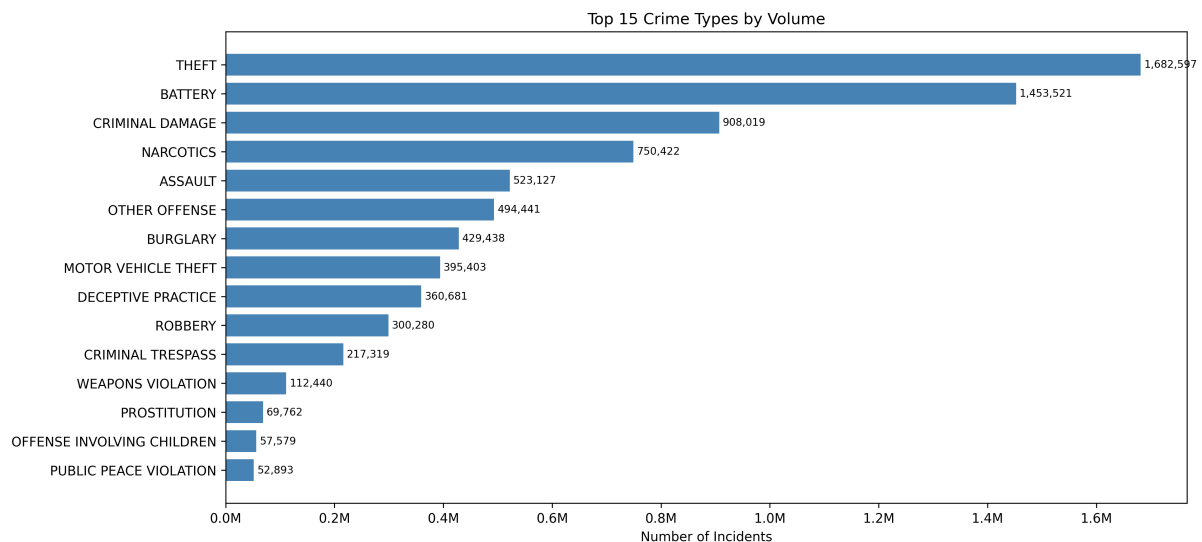


Figure A.2: Top 15 Crime Types by Volume

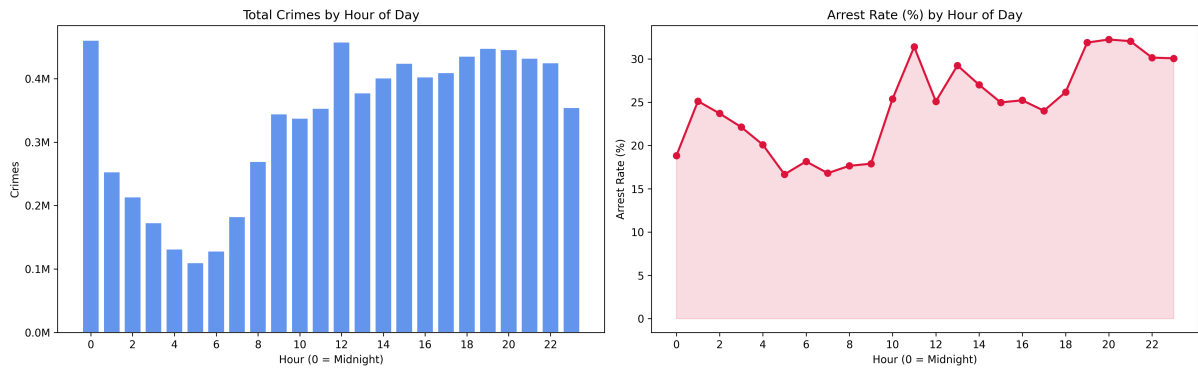


Figure A.3: Crimes and Arrest Rate by Hour

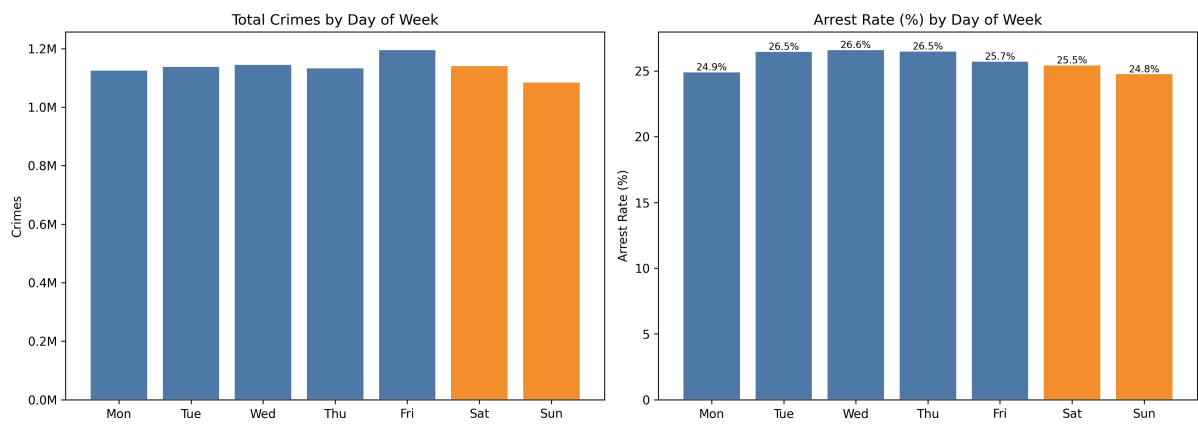


Figure A.4: Crimes and Arrest Rate by Day of Week

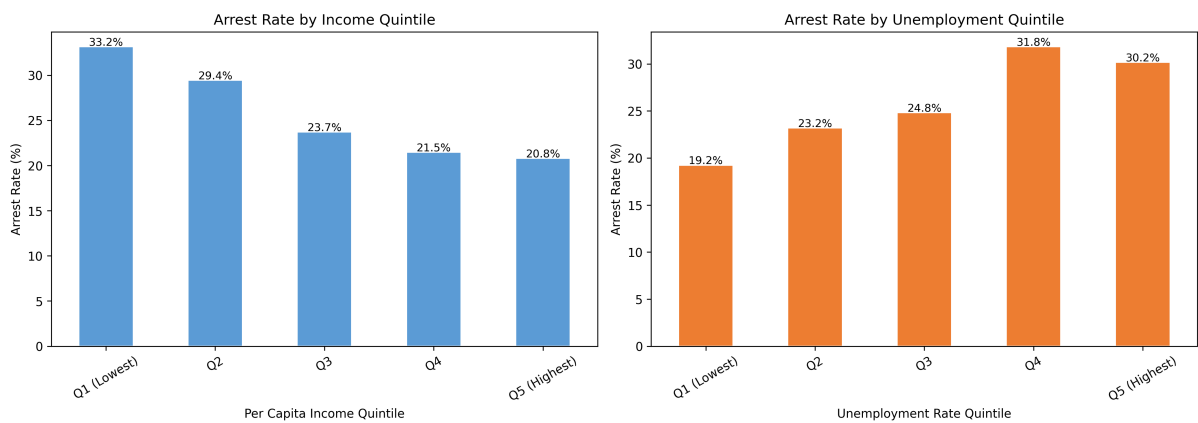


Figure A.5: Arrest Rate by Income and Unemployment Quintiles

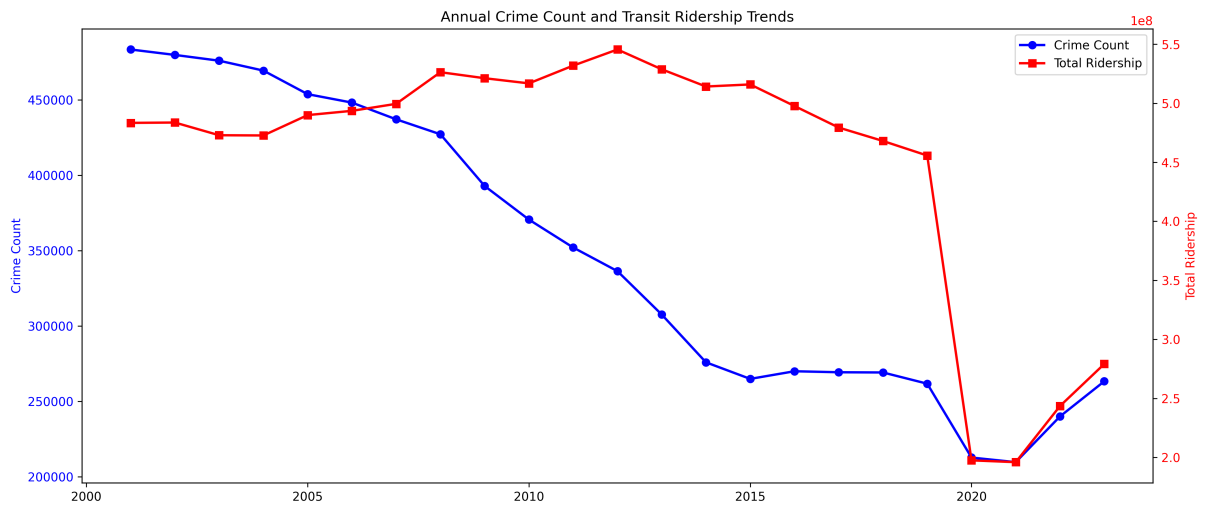


Figure A.6: Annual Crime Count and Transit Ridership Trends.

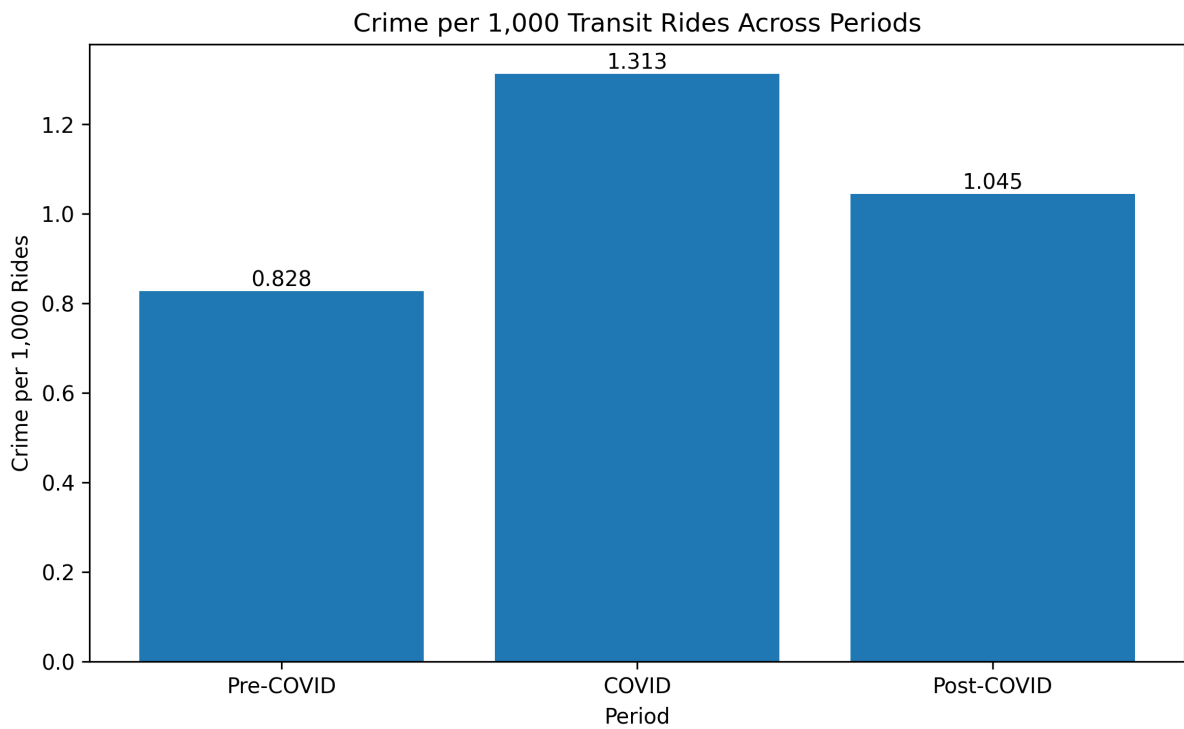


Figure A.7: Pre-COVID-19, COVID-19, and Post-COVID-19 Crime and Mobility Comparison

B. Supplementary Model Comparison Figures

This appendix presents the Accuracy, Precision, and Recall comparison charts across all evaluated models and sampling strategies. These are supplementary to the F1-score comparison in Chapter 5 (Figure 5.3): each shows the same overall ranking pattern viewed through a single metric, so only F1-score — the most representative single measure for this imbalanced classification task — is retained in the main text. These figures are provided here for readers who wish to inspect each metric individually.

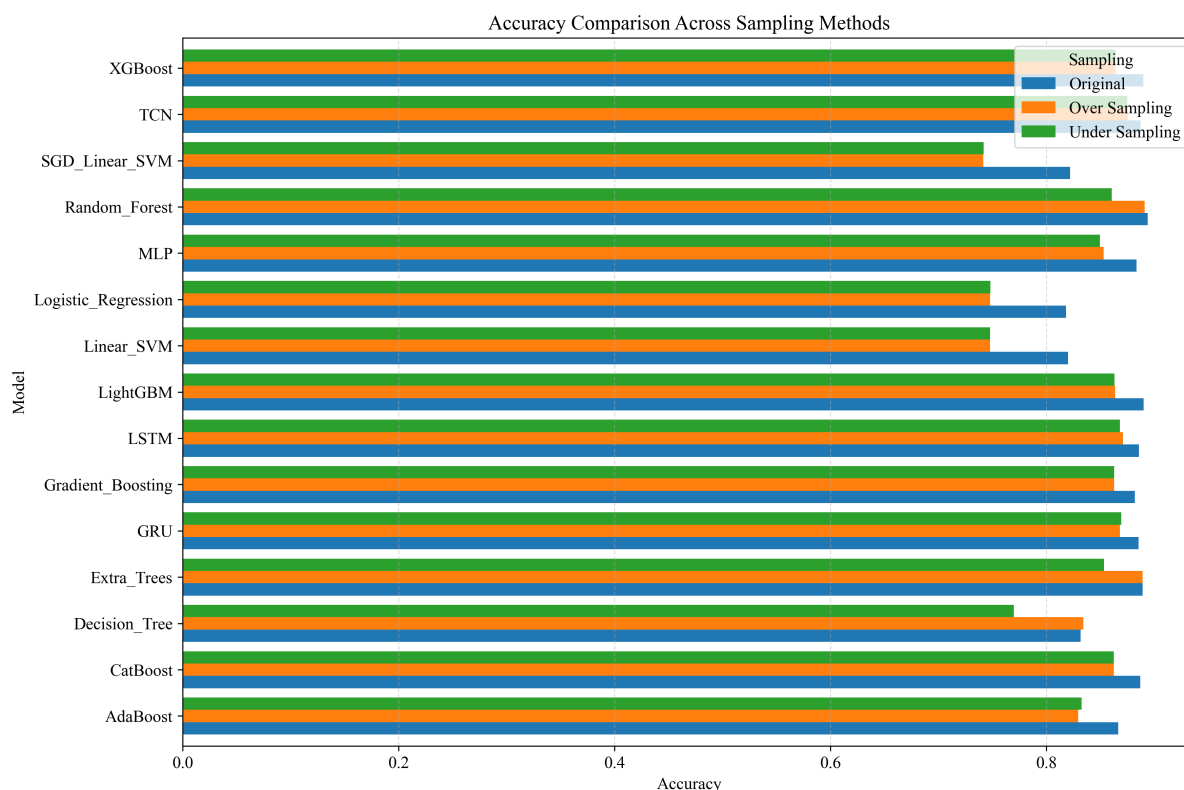


Figure B.1: Accuracy Comparison Across Sampling Methods

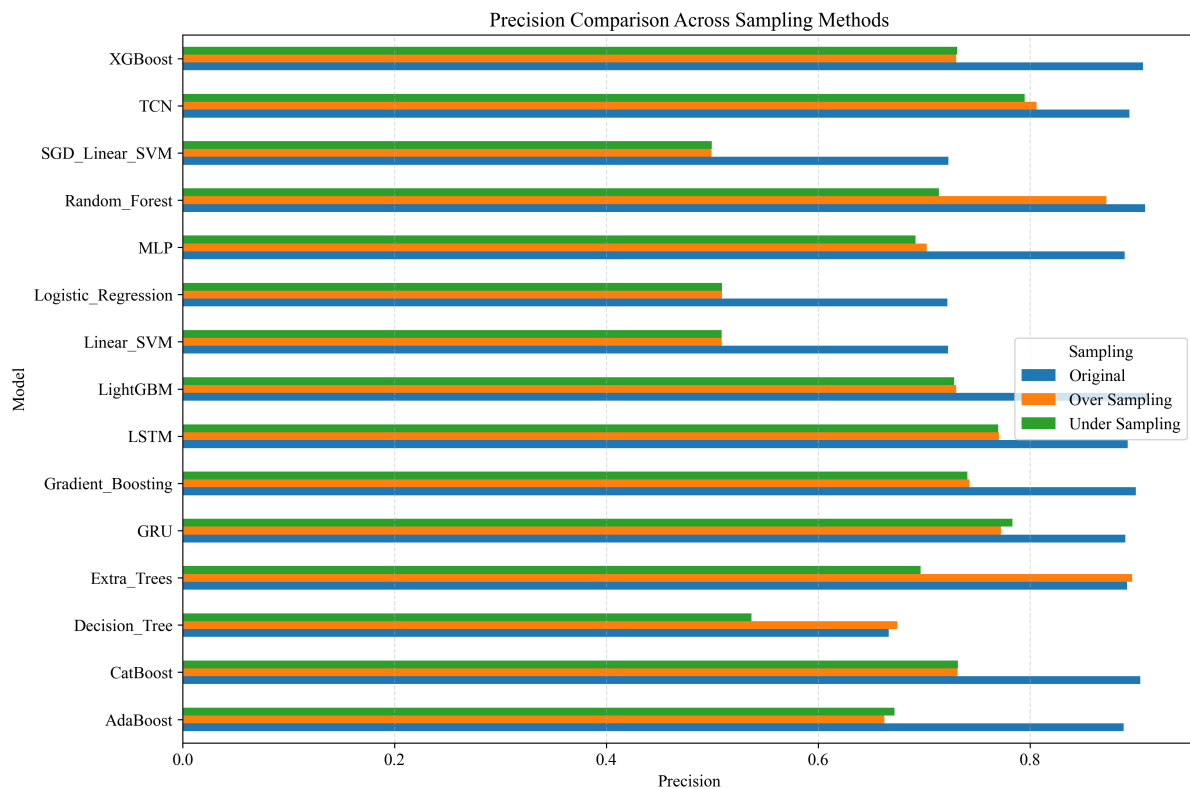


Figure B.2: Precision Comparison Across Sampling Methods

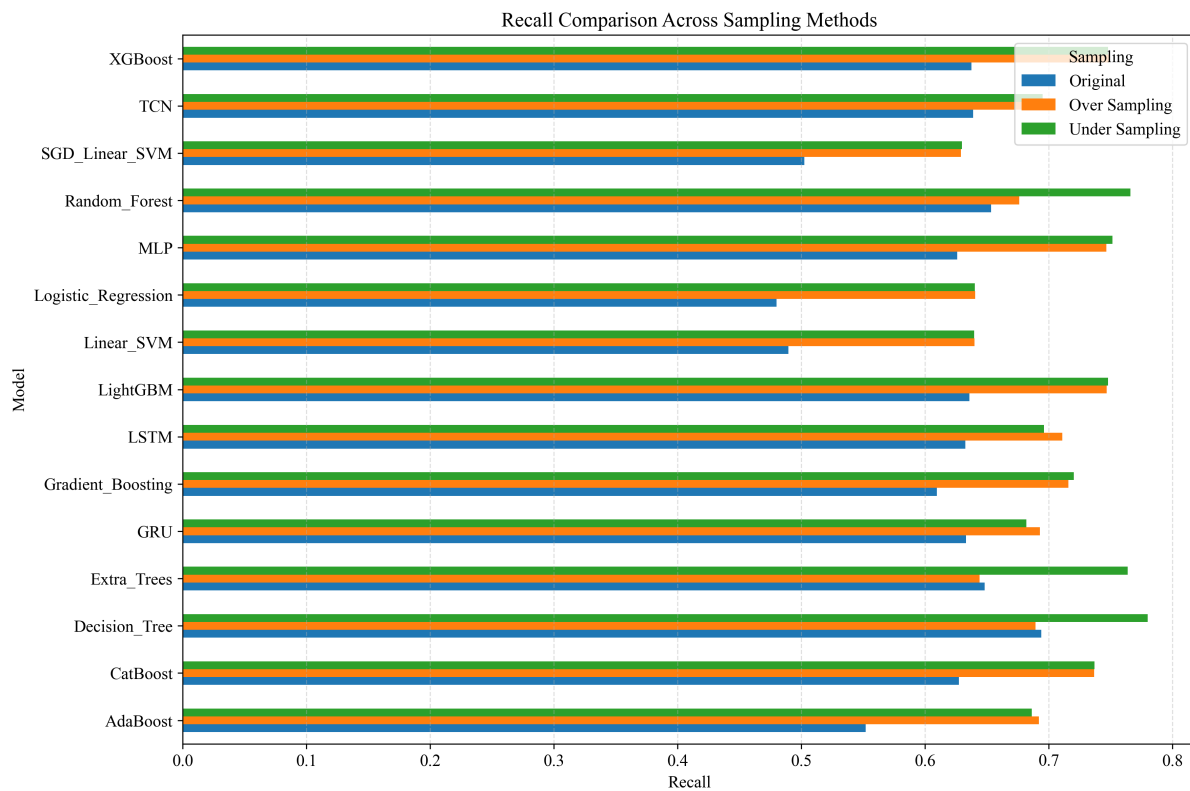


Figure B.3: Recall Comparison Across Sampling Methods

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